

Run Characteristics of Fall Chum Salmon Returns in the Upper Yukon River Basin

John H. Eiler and Jerome J. Pella

National Marine Fisheries Service, Auke Bay Laboratory

11305 Glacier Highway, Juneau, Alaska 99801

Randy J. Brown and Tevis J. Underwood

U.S. Fish and Wildlife Service, Fairbanks Fishery Resource Office

101 12th Avenue, Room 222, Fairbanks, Alaska 99701

Executive Summary

Fall chum salmon returning to spawning areas in the upper Yukon River basin were radio-tagged and tracked upriver in 1998 and 1999. This study was conducted in tandem with a mark-recapture program conducted by the U.S. Fish and Wildlife Service, which provided total and temporally-stratified run abundance estimates for the upper basin. The primary objective of the radio tagging study was to provide information on run characteristics, including movement patterns, stock composition and timing. Numerous research projects in the basin capture fish with fish wheel, often holding them in live boxes prior to being handled and released. A secondary objective of this study was to assess the potential impacts of holding fish in fish wheel live boxes.

Run timing was substantially different between the two years of the study, with most fish returning late in the run (late August to mid September) in 1998, while the peak of the 1999 run was in mid August. Fish were tagged during the last four weeks of the run (87.5% of the return) in 1998, while the entire return was sampled in 1999. The returns in both years were comprised primarily of fish returning to the Chandalar River, Sheenjek River and Yukon River in Canada. Smaller numbers of fish traveled to areas in the Porcupine River drainage, including the Black River, Fishing Branch River, and sections of the lower and upper Porcupine River. Fish also remained in the Yukon Flats, and were located in both secondary tributaries and areas associated with the Yukon River main stem. Chandalar River fish were the dominant stock during both years of the study, and were prevalent throughout the run. Sheenjek River fish were abundant in 1998, but declined substantially in 1999. The proportion tracked to the Yukon River in Canada was comparable during both years of the study, and was comprised of fish traveling to the Kluane River and other section of the upper Yukon River. Kluane River fish were primarily early run fish, while upper Yukon River fish exhibited a

later distribution. Although small numbers of fish remained in the Yukon Flats in 1998, this component increased significantly in 1999. However, these data are difficult to interpret due to the nature of the Yukon Flats (i.e., large, remote, and extremely turbid in most reaches), which makes it difficult to determine the status of the fish. In addition to returning to spawning areas in main stem reaches or associated tributaries, fish remaining in the Yukon Flats could represent fish that died while in-transit to upriver reaches due to handling at the tagging site, predation, or encounters with fishing gear; or a failure of the telemetry equipment used. The data supporting or contradicting these possible explanations are discussed.

Fish moved significantly faster on average in 1998 (40.0 km/day) than in 1999 (33.5 km/day). The limited information available suggests that water levels in the Yukon River main stem were higher in 1999, which may have affected movements. Although slower than in 1998, movement rates in 1999 were comparable to those reported in earlier studies in the Yukon River basin, and greater than other watersheds. Males and females exhibited similar movement rates during both years of the study, however, stocks returning to upper reaches of the basin moved significantly faster than fish returning to spawning areas lower in the basin.

Fish wheels have been used to study salmon returns in numerous rivers, providing an efficient method for capturing fish for enumeration, tagging and sampling projects. Live boxes are routinely used to hold fish until they are processed, reducing the time personnel need to be present at the site. Results from this study indicate that the behavior and movements exhibited by radio-tagged chum salmon held in fish wheel live boxes were significantly different compared to fish tagged and released immediately after capture. Held fish took significantly longer to resume upriver movements

and traveled shorter distances on average than immediately released fish, suggesting that holding fish in live boxes had an adverse impact on fish performance and survival.

Introduction

Large numbers of chum salmon (*Oncorhynchus keta*) return to the Yukon River basin to spawn. These returns consist of two fairly distinct runs, commonly referred to as summer and fall chum salmon (Gilbert 1922). Summer chum salmon spawn primarily in the lower and middle sections of the basin (the Tanana River and reaches downstream of the Yukon-Tanana River confluence), while fall chum salmon return to spawning areas in the Tanana River and the upper Yukon River basin above the Yukon-Tanana River confluence (Figure 1). Fall chum salmon returning to the upper basin include stocks destined for Canada (Anon 1985). The two runs are also separated temporally. Summer chum salmon first enter the lower river in early June with peak abundance in mid June, while fall chum salmon are first observed in the lower river in mid July and peak in early August.

Yukon River fall chum salmon support important commercial and subsistence fisheries in both the U.S. and Canada. These fisheries have been the focus of numerous disputes over harvest allocations between the two countries. Most fishing is concentrated near villages along the Yukon River main stem. Fish are also harvested in the lower reaches of the Chandalar, Porcupine, Stewart and Pelly Rivers, and near the village of Old Crow on the Porcupine River (Figure 1). Annual harvests of fall chum salmon have averaged around 276,000 fish in the U.S. and 20,000 fish in Canada since the early 1960s (Yukon River Joint Technical Committee 2000), with catches ranging from 62,869 fish (1998) to 654,976 fish (1981) in the U.S. and 3,711 fish (1970) to 45,600 fish (1995) in Canada. Detailed information on fall chum salmon returns, including run abundance, migratory patterns,

stock composition and timing, is needed to improve management of the fisheries and facilitate conservation efforts. However, the upper basin is large, remote, and extremely turbid in many areas, making this information difficult to obtain.

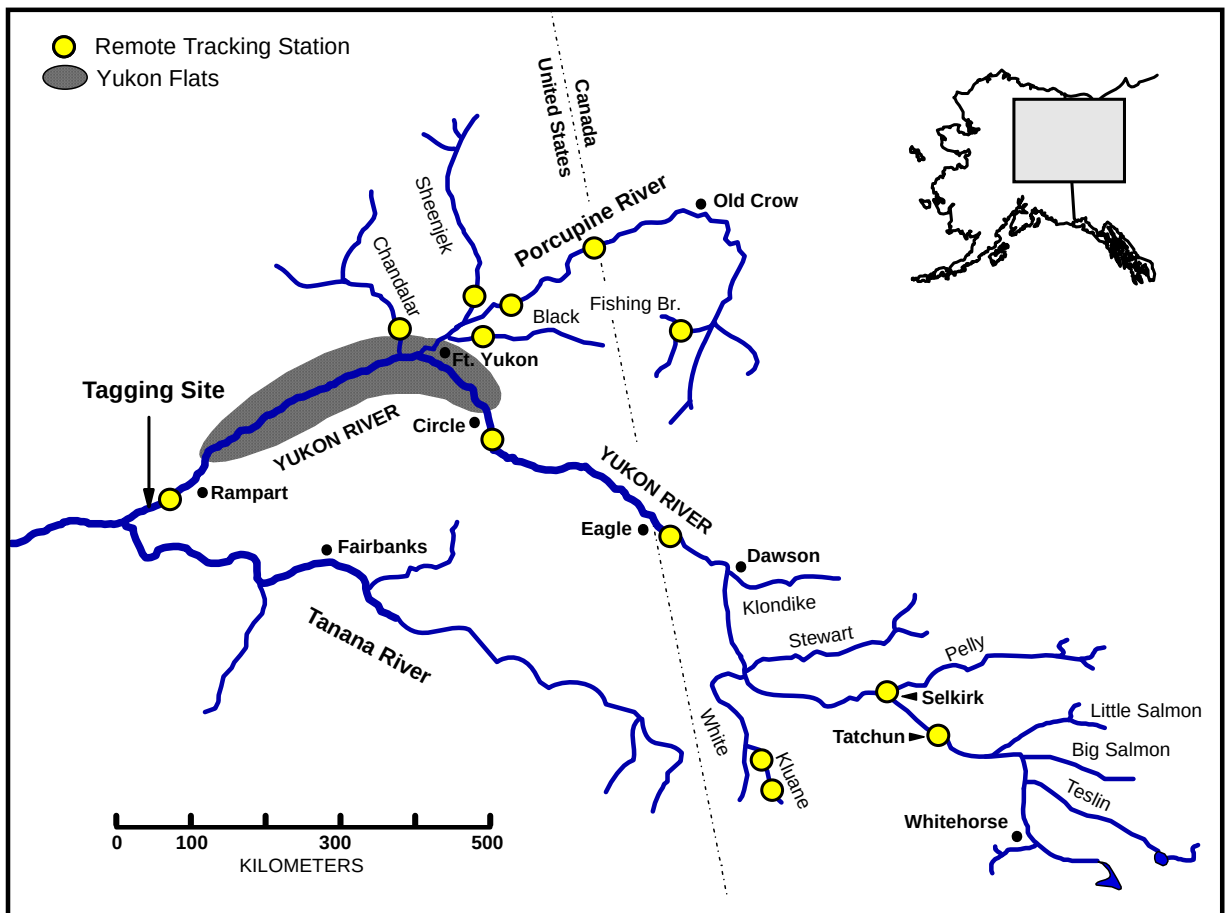


Figure 1. Map of the Tanana River and the upper Yukon River basin showing the Yukon River main stem and the major tributaries of the drainage. The tagging site and the location of remote tracking stations are also shown.

A cooperative, interagency tagging program was conducted from 1996 to 1999 to determine run characteristics for fall chum salmon returns in the upper Yukon River basin. The program consisted of two related components: radio tagging conducted by the National Marine Fisheries Service

(NMFS), and conventional tagging to develop mark-recapture estimates conducted by the U.S. Fish and Wildlife Service (USFWS). The conventional tagging and associated recapture efforts provided total and temporally-stratified run abundance estimates (Gordon et al. 1998, Underwood et al. 2000a, Underwood et al. 2000b). The objective of the radio tagging study was to provide information on spawner distribution, nation-of-origin, stock composition and timing, movement patterns, and the location of previously undocumented spawning areas. Preliminary work was conducted in 1996-1997 to determine capture and tagging feasibility, and to develop the infrastructure necessary for tracking radio-tagged fish in the upper basin. Large-scale radio tagging programs were conducted in 1998 and 1999.

Study Area

The upper Yukon River basin drains a watershed of about 500,500 km². The main river alone flows for over 1,700 km from its headwaters in Canada to the Tanana-Yukon River confluence. Several major tributaries flow into the Yukon River main stem, including the Chandalar, Porcupine, Stewart White, and Pelly Rivers. Most reaches of the drainage consist of a primary river channel with occasional side channels and sloughs, although the Yukon River main stem is extensively braided between the villages of Rampart and Circle – an area commonly referred to as the Yukon Flats (Figure 1). Sections of the White River are also extensively braided. Water visibility is extremely poor in many areas of the basin, particularly in the Yukon River main stem, due to turbidity from glacial runoff in the upper reaches of the drainage.

Methods

Fish Capture

Adult chum salmon returning to spawning areas in the upper basin were captured with fish wheels (Meehan 1960, Gordon et al. 1998) located 60 km upriver from the Yukon-Tanana River confluence (Figure 1). This site, known locally as the Rapids, was selected to 1) avoid sampling summer chum salmon or fall chum salmon destined for the Tanana River, 2) because the fish wheels, which were operated by local fishers, were an effective capture method even during years with poor chum salmon returns, and 3) because the site consisted of a single, relatively narrow (i.e., about 500 m width) channel, increasing the probability of capturing a representative sample of the return. The fish wheels were operated from August 3 to September 19, 1998, and August 4 to September 22, 1999 (Table 1).

Table 1. Schedule of fish wheel operations to capture fall chum salmon in the upper Yukon River basin in 1998 and 1999.

Capture Week	1998	1999
32	August 2-8	August 1-7
33	August 9-15	August 8-14
34	August 16-22	August 15-21
35	August 23-29	August 22-28
36	August 30-September 5	August 29-September 4
37	September 6-12	September 5-11
38	September 13-19	September 12-18
39		September 19-25

Tagging Procedure

Immediately after capture, the fish were placed in a neoprene-lined tagging cradle submerged in a trough of fresh water (Figure 2). Anesthesia was not used during the tagging procedure. The fish were tagged with a pulse-coded radio transmitter (Eiler 1995) in the 150-151 MHz frequency range. The transmitters – which were 5.1 cm long, 2.0 cm in diameter, had a 30 cm transmitting antenna, and weighed 20 g – were gently inserted through the mouth and into the stomach using a plastic tube 0.7 cm in diameter. Each transmitter emitted a unique signal (Eiler 1995), making it possible to identify individual fish, and was equipped with a motion sensor and activity monitor (Eiler 1990). The motion sensor, a small mercury switch sensitive to movement, inserted additional signal pulses distinct from the basic signal pattern each time the transmitter moved. The activity monitor altered the signal pattern to an inactive mode if the motion sensor was not triggered for 24 h; the signal reverted to the original pattern if the motion sensor was activated.



Figure 2. (Left) Radio-tagged salmon in the tagging cradle prior to release; the transmitter antenna and external spaghetti tag are visible. (Right) A necropsy of a radio-tagged salmon recovered on the spawning grounds showing the relative size and placement of the transmitter in the stomach.

The fish were marked externally with colored spaghetti tags attached below the dorsal fin (Wydoski and Emery 1983). Information on sex, length (mid eye to fork-of-tail), and condition of the fish was also recorded. A tissue sample was taken from the pelvic fin in 1999 for genetic stock identification analysis. The fish were released back into the main river immediately after the tagging procedure was completed. Handling, from capture by the fish wheels to release, took approximately five minutes.

Radio Tracking

Radio-tagged fish that moved upriver were tracked with remote tracking stations (Figure 3) located at sites throughout the upper basin (Figure 1). The sites selected were on important travel corridors and major tributaries of the drainage. When possible, the stations were placed on bluffs overlooking straight, narrow, single-channel sections of the river to maximize reception range and increase the probability of detecting fish moving past the site. The stations consisted of several integrated components, including a computer-controlled receiver developed by Advanced Telemetry Systems (Isanti, Minnesota), satellite uplink, and self-contained power system similar to that described by Eiler (1995). Radio-tagged fish within reception range were identified and recorded by the stations. The information collected included the date and time the fish were present at the site, the signal strength of the transmitter, and the orientation of the fish in relation to the station (i.e., upriver or downriver from the site). Because of the isolated nature of the sites, the telemetry data collected, including information on station operations (e.g., voltage levels for the station components, and whether the reference transmitter at the site was properly recorded) were transmitted every hour to a geostationary operational environmental satellite (GOES) and relayed to a receiving station near

Washington D.C. (Figure 3). The information was accessed daily via telephone modem, and the data files downloaded into a computerized data base for analysis (Eiler and Masters 2000).

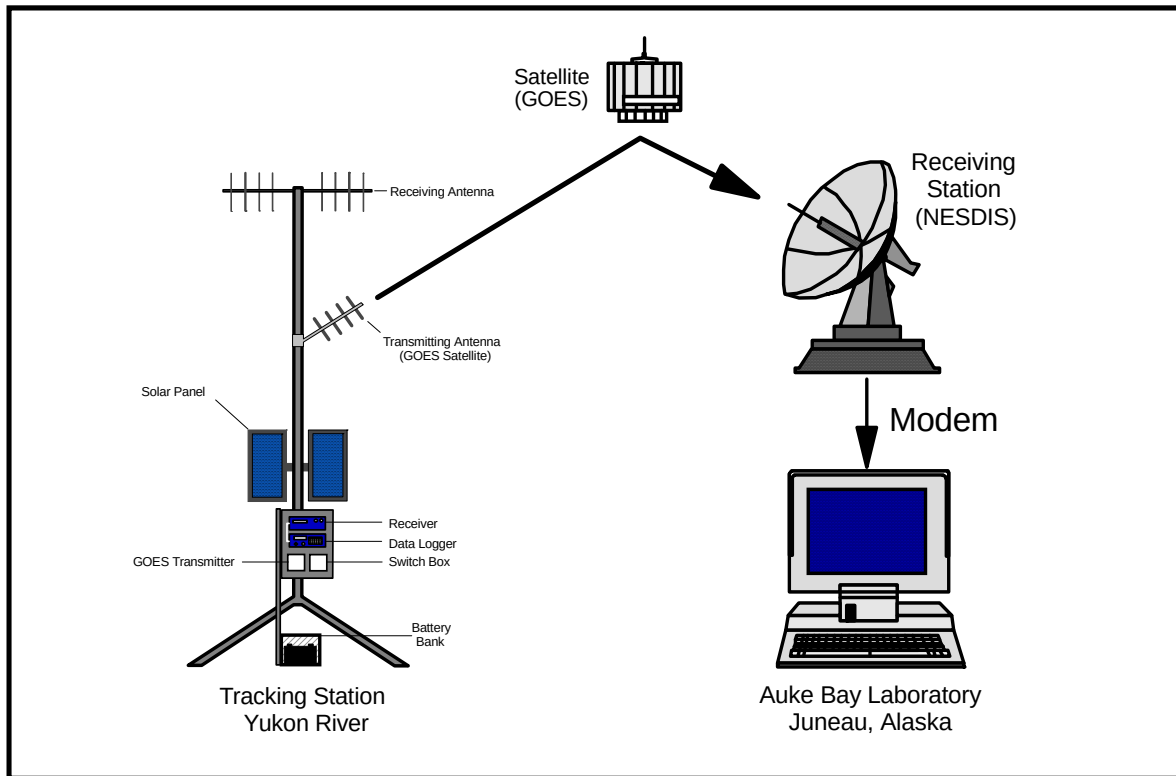


Figure 3. Remote tracking station and satellite uplink used to collect and access movement information for radio-tagged chum salmon in the upper Yukon River basin.

Aerial surveys were conducted in selected reaches of the drainage to locate radio-tagged fish that traveled to areas between station sites, and upriver of stations on terminal tributaries. The fish were tracked from fixed-wing aircraft and helicopters equipped with 4-element yagi receiving antennas placed on both sides of the aircraft and oriented forward. Tracking receivers were linked to a global positioning system (GPS) receiver to assist in identifying and recording locations. Limited tracking was also conducted by boat near the tagging site to document the movements of radio-tagged fish immediately after release.

Radio-tagged chum salmon that passed the first set of tracking stations, located 11 km upriver from the tagging site (hereafter referred to as the Rapids tracking stations), were considered to have resumed upriver movements. Fish tracked to terminal reaches of the drainage were classified as distinct spawning stocks. The status of fish that remained in non-terminal areas, such as the Yukon Flats, was less certain since these fish could represent local spawners or fish in-transit to spawning areas in the upper reaches of the drainage. Many non-terminal areas were turbid and hard to access, making verification of spawning activity difficult.

Stock Composition Estimation

Weekly abundance estimates of fall chum salmon passing the tagging site, derived from the USFWS mark-recapture program (Underwood et al. 2000), were allocated to spawning stocks based on the distribution of radio-tagged fish. Because subsistence and commercial catches upriver from the tagging site alter the initial stock composition estimates, the observed weekly distributions of radio-tagged fish among spawning areas were adjusted to account for fish removed from the sample. Adjustments were based on a migration model; radio-tagged fish migrating upstream from the tagging site during each capture week were modeled to reflect the geographical branching of the primary upriver tributaries (Figure 1) and associated spawning stocks. At the lowest branching point above the tagging site, stocks and the stock proportions included the Chandalar River, p_1 ; Yukon Flats, p_2 ; and remaining stocks further upriver, p_3 ($p_1 + p_2 + p_3 = 1$). Radio-tagged fish caught in the Fort Yukon fishery were assumed to be from spawning populations further upriver. At the second branching point, beyond the Fort Yukon fishery, radio-tagged fish separated into the Porcupine River drainage (proportion q_1) or Yukon River drainage (proportion q_2) ($q_1 + q_2 = 1$). Within the Porcupine River drainage, radio-tagged fish branched into the Sheenjek River (proportion $r_{1(1)}$), the

U.S. portion of the drainage hereafter referred to as the lower Porcupine River (proportion $r_{2(1)}$), Black River (proportion $r_{3(1)}$), or the Canadian portion of the drainage (proportion $r_{4(1)}$) ($r_{1(1)}+r_{2(1)}+r_{3(1)}+r_{4(1)}=1$). At the final branching point, upstream from the U.S.-Canada border, radio-tagged fish destined for the upper Porcupine River (i.e., areas other than the Fishing Branch River)(proportion $s_{1(1)}$) or Fishing Branch River (proportion $s_{2(1)}$) ($s_{1(1)}+s_{2(1)}=1$) were caught in the Old Crow fishery. Beyond the Fort Yukon fishery, radio-tagged fish that migrated up the Yukon River encountered the Circle fishery, and then either remained in the area between Circle and Eagle (proportion $r_{1(2)}$) or continued upriver ($r_{2(2)}$) ($r_{1(2)}+r_{2(2)}=1$). Fish continuing upriver encountered fisheries at Eagle and Dawson before completing their migration to the Canadian portion of the Yukon River downstream from the Selkirk tracking station (proportion $s_{1(2)}$), the Kluane River (proportion $s_{2(2)}$), the area between the Selkirk and Tatchun tracking stations, including the Pelly River (proportion $s_{3(2)}$) or the area above the Tatchun tracking station (proportion $s_{4(2)}$) ($s_{1(2)}+s_{2(2)}+s_{3(2)}+s_{4(2)}=1$). Fish returning to areas other than the Kluane River are hereafter referred to as upper Yukon River fish.

Counts of the weekly releases of radio-tagged fish caught in fisheries and escaping to spawning areas are associated with the branching proportions. The total number of radio-tagged fish moving upriver from the tagging site, T , equals the sum of 1) the number to the Chandalar fish, X_1 , 2) the number staying in the Yukon Flats, X_2 , and 3) the numbers caught, C_3 , or passing further upriver, X_3 , at Fort Yukon. The proportions to the corresponding branches were estimated as:

$$\begin{aligned}\hat{p}_1 &= X_1 / T \\ \hat{p}_2 &= X_2 / T \\ \hat{p}_3 &= (X_3 + C_3) / T\end{aligned}$$

where $T = X_1 + X_2 + C_3 + X_3$.

The numbers passing upriver from the Fort Yukon branch to the Porcupine River and Yukon River drainages, the combined total numbers to the Porcupine River, T_1 , and the Yukon River, T_2 , were used to estimate the branching proportions to the drainages:

$$\hat{q}_i = T_i / (T_1 + T_2), \quad i = 1, 2.$$

The total numbers to the Porcupine River drainage, T_1 , were composed of the numbers escaping to the Sheenjek River, $Y_{1(1)}$, the lower Porcupine River, $Y_{2(1)}$, the Black River, $Y_{3(1)}$, and the Canadian section of the drainage, $Y_{4(1)}$. The numbers passing further upriver into Canada included the radio-tagged fish caught in the Old Crow fishery, $C_{1(1)}$, and those escaping to the upper Porcupine River, $Z_{1(1)}$, and Fishing Branch River, $Z_{2(1)}$. These counts were used to estimate the branching proportions in the Porcupine River drainage:

$$\begin{aligned} \hat{r}_{i(1)} &= Y_{i(1)} / T_1, \quad i = 1, 2, 3, 4, \quad \text{where } T_1 = \sum_{i=1}^4 Y_{i(1)} \\ \hat{s}_{i(1)} &= Z_{i(1)} / (Z_{1(1)} + Z_{2(1)}), \quad i = 1, 2 \end{aligned}$$

The total numbers upriver of Fort Yukon that branch to the Yukon River, T_2 , were composed of the radio-tagged fish (1) caught in the Circle fishery, $C_{1(2)}$, (2) remaining between Circle and Eagle, $Y_{1(2)}$, and (3) passing further upriver, $Y_{2(2)}$. Of those passing further upriver, some were caught in fisheries at Eagle and Dawson, $C_{2(2)}$, and the remainder escaped to the Canadian portion of the Yukon River downstream from the Selkirk tracking station, $Z_{1(2)}$, the Kluane River, $Z_{2(2)}$, the area between the Selkirk and Tatchun tracking stations, $Z_{3(2)}$, and the area above the Tatchun tracking station, $Z_{4(2)}$. These counts to the Yukon River were used to estimate the branching proportions:

$$\hat{r}_{i(2)} = Y_{i(2)} / T_2, \quad i = 1, 2, \quad \text{where } T_2 = Y_{1(2)} + Y_{2(2)}$$

$$\hat{s}_{i(2)} = Z_{i(2)} / \sum_{i=1}^4 Z_{i(2)}, \quad i = 1, 2, 3, 4.$$

Weekly estimates of proportions at the tagging site bound for any stock were computed as the product of corresponding weekly estimates of branching proportions leading to the stock. For example, the estimated proportion bound for the Chandalar River is $\hat{p}_1$, for the Black River is $\hat{p}_1 \hat{q}_1 \hat{r}_{3(1)}$, and for the Fishing Branch River is $\hat{p}_1 \hat{q}_1 \hat{r}_{4(1)} \hat{s}_{2(1)}$.

Weekly estimates of numbers of each stock passing the tagging site were computed by multiplying the spaghetti-tagging estimates of total weekly abundance by the estimated weekly stock proportions from radio tagging. Variances of these weekly numbers of migrants were computed using Goodman's (1960) formula for the variance of products, with the weekly variances for total abundance from the spaghetti-tagging program and approximate binomial variances for the estimated weekly stock proportions from the radio-tagging program. Season totals of numbers past the tagging site for stocks were computed by summing their weekly abundance estimates; and variances of these season totals were the sums of the variances of the comprising weekly abundance estimates. The estimated season total arriving at the spawning areas of any stock was obtained by subtracting allocated season catches in the fisheries from season totals past the tagging site; weekly catches were unavailable so seasonal allocation was necessary. Season catches in fisheries were allocated in proportion to estimated season numbers to stocks upriver from the fishery.

Holding Experiments

Fish wheels used to catch and release salmon for run enumeration and other research projects are often equipped with live boxes (Meehan 1960) to hold the fish prior to handling. Experiments were conducted during this study to evaluate the effects of holding chum salmon in fish wheel live boxes prior to being tagged and released. The holding boxes used at the tagging site were approximately 2.4 m long, 1 m wide and 1.2 m deep (Gordon et al. 1998). The boxes were constructed to allow a continuous flow of river water through the structure while eliminating strong currents that could injure the fish.

Fifty chum salmon were radio tagged during a 6-day period from August 22-27, 1996. Twenty-five fish were tagged immediately after capture with radio transmitters and spaghetti tags and released (i.e., standard handling procedure). The other twenty-five fish were held in the fish wheel live box for 4-6 hours prior to being tagged and released. The time taken by the fish to resume upriver movements and pass the Rapids tracking stations was recorded. Fish recaptured by the fish wheels and released were excluded from the analysis.

In 1998, three different handling procedures were used on chum salmon. The fish were either tagged using the standard handling procedure, tagged immediately after capture with only radio transmitters and released, or held in fish wheel live boxes for 4-6 hours prior to being radio-tagged, spaghetti-tagged and released. Each treatment group consisted of 80 fish. The held fish were selected from a 30-fish sample confined in the holding box. Approximately half of the holding box was cordoned off to isolate fish held for the experiment. The test was conducted during a 7-day period from August 25-31 with comparable numbers of fish tagged each day using the different handling

procedures. The time taken by the fish to resume upriver movements and pass the Rapids tracking stations was recorded. The fish were also tracked upriver to compare the distribution and movement patterns of the three treatment groups. Radio-tagged fish, recaptured and released at fish wheels at the tagging site or the village of Rampart (50 km upriver), were excluded from the analysis.

In 1999, 100 fish captured during a 10-day period from 25 August to 4 September were held in the fish wheel holding box prior to being tagged and released. These fish were separated into two treatment groups: 50 fish were held for 4-6 hours under crowded conditions (i.e., fish tagged that day selected from a 45-fish sample), and 50 fish were held for 4-6 hours under less crowded conditions (i.e., selected from a 10-fish sample) prior to being tagged and released. Approximately half of the holding box was cordoned off to isolate fish held for the experiment. The time taken by the fish to resume upriver movements and pass the Rapids tracking stations was recorded. The movements and distribution of the fish from these two treatment groups were compared with fish tagged during the same period using the standard handling protocol. Radio-tagged fish, recaptured by fish wheels at the tagging site or Rampart, were excluded from the analysis.

Ancillary data from 1999 was also used to evaluate the impacts of holding on fish performance. The distribution of tagged fish that moved upriver without being recaptured was compared with fish caught and released a second time at the fish wheels at Rampart. No information is available on the length of time these fish were confined. The fish wheels were operated 24 hours per day, and the live box checked two to four times per day, so that holding times may have ranged from one to twelve hours. Efforts were made to limit the fish in the live box to less than 200 fish, but this number was periodically exceeded (Gordon et al. 1998). Insufficient data was obtained in 1998 to make similar comparisons; only nine fish were reported as recaptures in the Rampart fish wheels.

Contingency table tests for goodness of fit (Raymond and Rousset 1995) were used to analyze differences observed between treatment groups, including sample percentages that moved upriver and proportional distribution in upriver areas. Significance levels were designated as $P < .05$. Differences in composition, travel times and movement rates between stocks and treatment groups in the holding experiment were evaluated using 95% confidence intervals (Sokal and Rohlf 1969). Three-way analysis of variance was used to compare movement rates by year, stock and gender using *Splus 2000 for Windows*.

Results

Tagging and Tracking Sample

A total of 1627 chum salmon were radio tagged during the study, including 1397 fish (451 fish in 1998 and 946 fish in 1999) tagged and released immediately after capture (Table 2). Fish tagged in 1996 or held in fish wheel live boxes during 1998 and 1999 as part of the holding experiment are discussed in a subsequent section. Most immediately released fish resumed upriver movements, including 428 (94.9%) fish in 1998 and 898 (94.9%) fish in 1999. The remaining fish either died due to handling or predation, regurgitated their tags, or traveled to other areas of the basin. Several radio-tagged fish that did not move upriver past the Rapids tracking station were later recovered in fish wheels located over 150 km up the Tanana River.

A substantial number of fish that moved upriver were recaptured and released by research fish wheels at the tagging site and the village of Rampart, including 52 fish in 1998 and 228 fish in 1999 (Table 3). This was particularly significant in 1999 when over 25% of the sample was recaptured. Distribution and movement data from these fish were excluded from the analyses due to the possible

effects from being held in fish wheel live boxes and the additional handling. Seventy-one fish were caught in the U.S. fishery including 16 (3.7%) fish in 1998 and 55 (6.1%) fish in 1999. With the exception of the fishery near Eagle, Alaska (Figure 1), these fish were likely destined for spawning areas in both the U.S. and Canada. Fish caught near Eagle were assumed to be Canadian stocks. Thirty-six fish were caught in Canada including 8 (1.9%) fish in 1998 and 28 (3.1%) fish in 1999. The remaining fish, 352 (82.2%) fish in 1998 and 587 (65.4%) fish in 1999, were tracked upriver without being caught in research fish wheels or local fisheries (Table 3), and were assumed to have reached their final destination.

Table 2. Numbers of chum salmon tagged with radio transmitters in the upper Yukon River basin by capture week and handling treatment during 1996, 1998, and 1999.

Capture Week	1996		1998		1999	
	Immediate Release	Held ¹	Immediate Release	Held ¹	Immediate Release	Held ¹
32					51	
33					102	
34					193	
35	25	25	76	32	205	40
36			76	38	110	60
37			171	10	131	
38			128		100	
39					54	
Total	25	25	451	80	946	100

¹ Held in fish wheels live boxes for 4-6 hours prior to being tagged and released.

Upriver Movements

Distribution of radio-tagged fish. Radio-tagged fish traveled to areas throughout the upper basin in 1998. Fish tracked to the Chandalar and Sheenjek Rivers comprised the largest components of the sample, with 126 (34.8%) fish and 97 (26.8%) fish returning to these two reaches, respectively (Table 4). Small numbers of fish were also tracked to other areas in the U.S., including the Black River (8, 2.2%), lower Porcupine River (7, 1.9%), and Yukon Flats (17, 4.7%). A total of 107 (29.6%) fish returned to the Canadian section of the basin, including 77 (21.3%) fish in the Yukon River (i.e., upper Yukon and Kluane River fish) and 30 (8.3%) fish in the Porcupine River (i.e., upper Porcupine and Fishing Branch River fish). Most (49, 13.5%) upper Yukon River fish remained downstream from the Selkirk tracking station (Figure 1). Small numbers of fish were also tracked to reaches between the Selkirk and Tatchun tracking stations (8, 2.2%), and upriver from the Tatchun tracking station (5, 1.4%). Ten (2.8%) fish traveled to the Kluane River. Five (1.4%) fish were caught in the U.S. fishery near Eagle and the Canadian fishery near Dawson (Figure 1); the stock composition of these fish is unknown. Nine (2.5%) fish returning to the Canadian portion of the Porcupine River were tracked to the Fishing Branch River, while 16 (4.4%) were upper Porcupine River fish. Five (1.4%) fish were caught in the Canadian fishery near Old Crow (Figure 1) and may represent fish destined for either area.

In 1999, Chandalar River fish again comprised the largest component of the sample, with 193 (30.6%) fish returning to this area (Table 4). Ninety-three (14.7%) fish traveled to the Sheenjek River. Small numbers of fish were tracked to the Black River (16, 2.5%) and the lower Porcupine River (6, 1.0%), while numerous (162, 25.7%) fish remained in the Yukon Flats. A total of 161 (25.5%) fish returned to the Canadian section of the basin, including 109 (17.3%) fish in the upper

Yukon River and 52 (8.2%) fish in the upper Porcupine River. Most (47, 7.4%) upper Yukon River fish remained downstream from the Selkirk tracking station. Small numbers of fish were also tracked to reaches between the Selkirk and Tatchun tracking stations (10, 1.6%), and upriver from the Tatchun tracking station (5, 0.8%). Eleven (1.7%) fish traveled to the Kluane River. A total of 36 (5.7%) fish were caught in the U.S. fishery near Eagle and the Canadian fishery. Most radio-tagged fish in the Canadian fishery were caught near Dawson, although some were also recovered near villages further upriver. Nineteen (3.0%) fish returning to the Canadian section of the Porcupine River were tracked to the Fishing Branch River, while 25 (4.0%) were upper Porcupine River fish. Eight (1.3%) fish were caught in the Canadian fishery near Old Crow and may represent fish destined for either area.

Table 3. Tracking results for immediately released radio-tagged chum salmon that resumed upriver movements in the upper Yukon River basin in 1998 and 1999. Percentages of the total are in parentheses.

Tracking Results	1998	1999
Recaptured in research fish wheels ¹	52 (12.2)	228 (25.4)
U.S. Fishery	16 (3.7)	55 (6.1)
Canadian Fishery	8 (1.9)	28 (3.1)
Upriver Location ²	352 (82.2)	587 (65.4)
Total	428	898

¹ Fish wheels located at the Rapids tagging site and the village of Rampart.

² Fish recorded past tracking stations upriver from the tagging site and not caught in fisheries.

Differences in the distribution of radio-tagged fish were observed in 1998 and 1999. The percentages

of fish returning to the Chandalar River, Black River, Porcupine River (both lower and upper sections), Fishing Branch River, and the Old Crow fishery were comparable during both years of the study (Table 4). However, a lower proportion of fish returned to the Sheenjek River in 1999 (14.7%) compared to 1998 (26.8%), while a considerably higher proportion of fish remained in the Yukon Flats in 1999 (25.7%) compared to 1998 (4.7%). Although the percentage of fish traveling to the upper Yukon and Kluane Rivers was similar in both years (21.3% in 1998 and 17.3% in 1999), the catch in the Canadian fishery was higher in 1999 and the fish tracked to areas further upriver correspondingly lower.

Limited aerial surveys were flown to locate radio-tagged fish in the Yukon Flats during mid November 1999 due to the large number of fish that remained in this section of the basin compared to the previous year. The areas covered included most sections of the Yukon River main stem, and several main stem tributaries including Minook Creek, Hess Creek, Big Salt River, Ray River, Dall River, Beaver Creek, and Hodzana River. Lower sections of the Charley, Kandik, Nation and Seventymile Rivers, located upriver from Circle, Alaska, were also surveyed. Eighteen (11.1% of the Yukon Flats component) fish were located in main stem tributaries (Table 4), including Hess Creek, Big Salt River, Ray River, Beaver Creek, Hodzana River, Charley River and the Nation River. Most (12) of these fish were tagged during the first week (Week 32). Thirty-five (21.6% of the Yukon Flats component) fish were located in the Yukon River main stem, and may represent fish that spawned in this area, died while in-transit to upriver areas due to handling, predation or injury in the fishery, or drifted downstream from other areas after spawning. Some of these fish were detected immediately downstream from main stem tributaries where radio-tagged fish had been located. A total of 109 (67.3% of the Yukon Flats component) fish that traveled to the Yukon Flats were not

located during the November surveys.

Table 4. Distribution of radio-tagged chum salmon in the upper Yukon River basin in 1998 and 1999, including fish caught in border and Canadian fisheries. Percentages of total are in parentheses.

Drainage	Final Location (Nation of Origin)	1998	1999
Yukon	Chandalar River (U.S.)	126 (34.8)	193 (30.6)
	Yukon River		
	Yukon Flats (U.S.) ¹	17 (4.7)	144 (22.8)
	Yukon Flats tributaries (U.S.) ²	--	18 (2.9)
	Upriver of U.S.-Canada Border (Can)	49 (13.5)	47 (7.4)
	Between Selkirk and Tatchun tracking station (Can)	8 (2.2)	10 (1.6)
	Upriver of Tatchun tracking station (Can)	5 (1.4)	5 (0.8)
	Kluane River (Can)	10 (2.8)	11 (1.7)
	Eagle, Alaska and Canadian Fishery (Can)	5 (1.4)	36 (5.7)
Porcupine	Sheenjek River (U.S.)	97 (26.8)	93 (14.7)
	Black River (U.S.)	8 (2.2)	16 (2.5)
	Porcupine River		
	Lower Porcupine River (U.S.)	7 (1.9)	6 (1.0)
	Upper Porcupine River (Can)	16 (4.4)	25 (4.0)
	Fishing Branch River (Can)	9 (2.5)	19 (3.0)
	Canadian Fishery (Can)	5 (1.4)	8 (1.3)
Total		362	631

¹ Fish that moved past remote tracking stations located 11 km upriver from the tagging site but were not recorded by tracking stations further upriver.

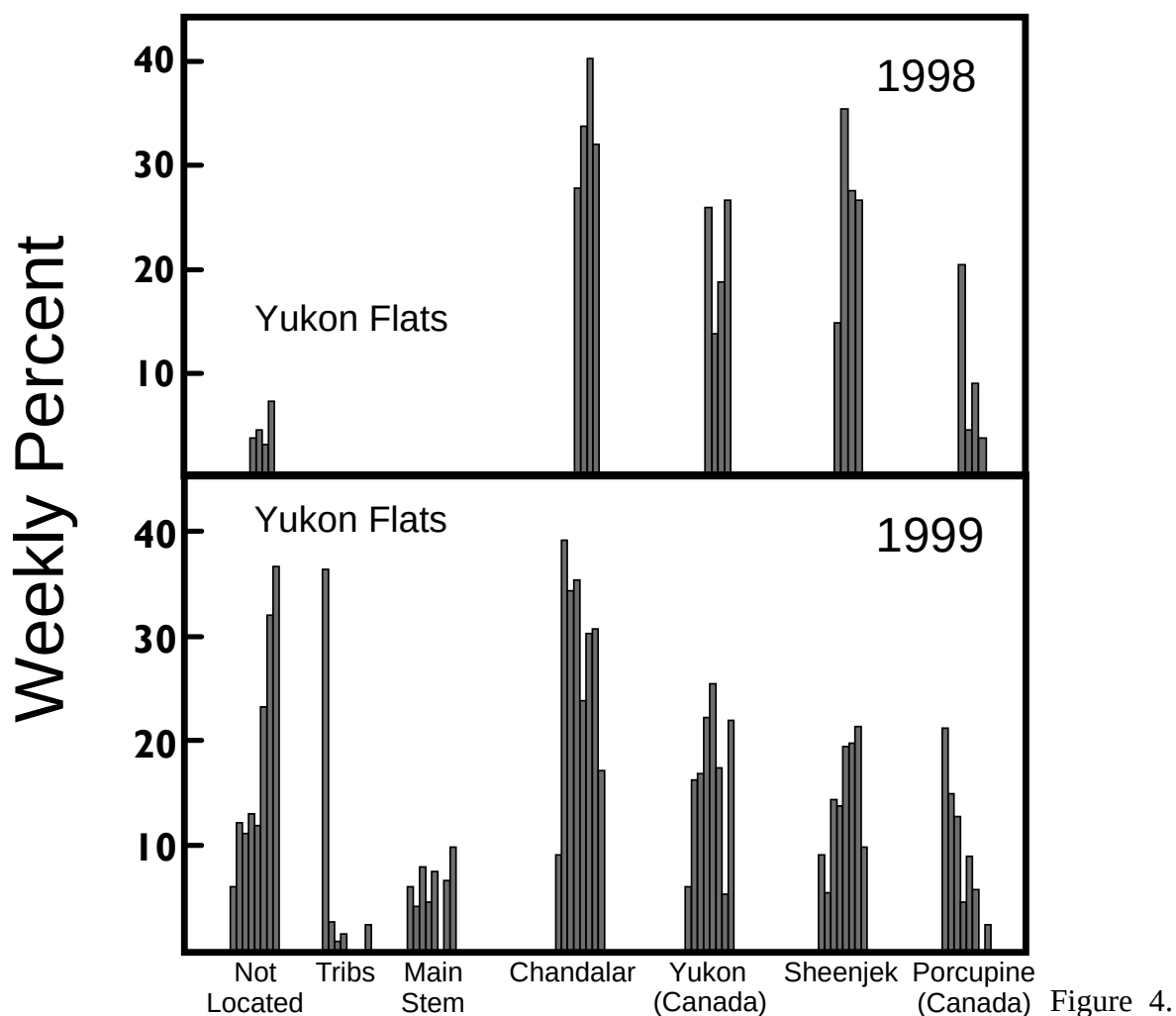
² Fish located in secondary tributaries of the Yukon River main stem during aerial surveys conducted in 1999; comparable surveys were not flown in 1998.

Weekly distribution of radio-tagged fish. Variations in timing were observed for radio-tagged fish

tracked to different reaches of the basin. Chandalar River fish were abundant at the capture site throughout the tagging period in 1998 (Week 35-38), ranging from 27.8% to 40.3% of the weekly samples (Figure 4). Sheenjek River fish were also abundant, particularly during the last three weeks of tagging when they comprised between 26.6% and 35.4% of the samples. The largest proportions of Yukon River fish in Canada were observed during the first (25.9%) and last (26.6%) week of tagging. This component was comprised of both Kluane and upper Yukon River fish during the first two weeks of tagging, while only upper Yukon River fish were present during the last two weeks of tagging. Fish returning to the Porcupine River in Canada were captured primarily during the first week of tagging. Small numbers of fish were tracked to the lower Porcupine and Black Rivers throughout the tagging period. The numbers of radio-tagged fish remaining in the Yukon Flats were minimal, with the greatest proportion observed during the last week of tagging (Figure 4).

Chum salmon were radio tagged during the entire run (Week 32-39) in 1999. Fish returning to the Chandalar River were abundant throughout the run, particularly during Week 33-38 when they comprised between 23.9% and 39.2% of the weekly samples (Figure 4). Sheenjek River fish were less prevalent than in 1998, ranging from 5.4% to 21.3% of the samples. The timing of Porcupine River fish in Canada was similar during both years of the study, with most fish tagged early in the run. Small numbers of fish were tracked to the lower Porcupine and Black Rivers throughout the tagging period. Yukon River fish in Canada were most abundant during the peak of the run, however relatively few fish tagged during Week 38 were tracked to this area. Analysis of station records from the U.S.-Canada border indicated that technical problems occurred during the period when fish tagged in Week 38 were moving past the site, reducing the percentage of fish designated to this area during this week. Fish tracked to the Kluane River were tagged primarily during the early portion

of the run, while upper Yukon River fish were present throughout the return. Substantially greater numbers of fish were allocated to the Yukon Flats during 1999, but the fall aerial surveys provided additional information about this component. A substantial proportion of the sample in Week 32 (32.4%), and smaller proportions in subsequent weeks, returned to small tributaries within this area. Fish tagged throughout the run were located in main stem areas, although these fish were more prevalent later in the run. This pattern was even more distinct for fish tracked to the Yukon Flats, but not located during fall surveys (Figure 4).



Weekly percentages of radio-tagged chum salmon tracked to reaches in the upper Yukon River basin in 1998 and 1999. Only a portion of the run (Week 35-38) was sampled in 1998, while the entire run (Week 32-39) was sampled in 1999.

Movement rates. Radio-tagged fish traveled an average of 40.0 km/day in 1998, ranging from 38.3 km/day for Sheenjek River fish to 43.7 km/day for fish traveling to the upper Yukon River. The fastest movement rate recorded was 54.9 km/day for a fish (male, 56 cm in length) that traveled past the Yukon River border, a distance of over 890 km. Chandalar, Sheenjek and Porcupine River fish exhibited comparable movement rates. No difference was observed between fish tracked to the lower and upper Porcupine River. Fish returning to the upper Yukon River, however, moved significantly faster than fish tracked to other reaches (Figure 5).

Radio-tagged fish traveled an average of 33.5 km/day in 1999, ranging from 30.5 km/day for Sheenjek River fish to 36.5 km/day for upper Yukon River fish. The fastest movement rate observed was 49.8 km/day for a fish (male, 58 cm in length) traveling past the Yukon River border. Similar movement rates were observed for Chandalar and Sheenjek River fish. Fish returning to the Porcupine River and upper Yukon River moved significantly faster, with migration rates of 36.5 km/day and 35.5 km/day, respectively (Figure 5).

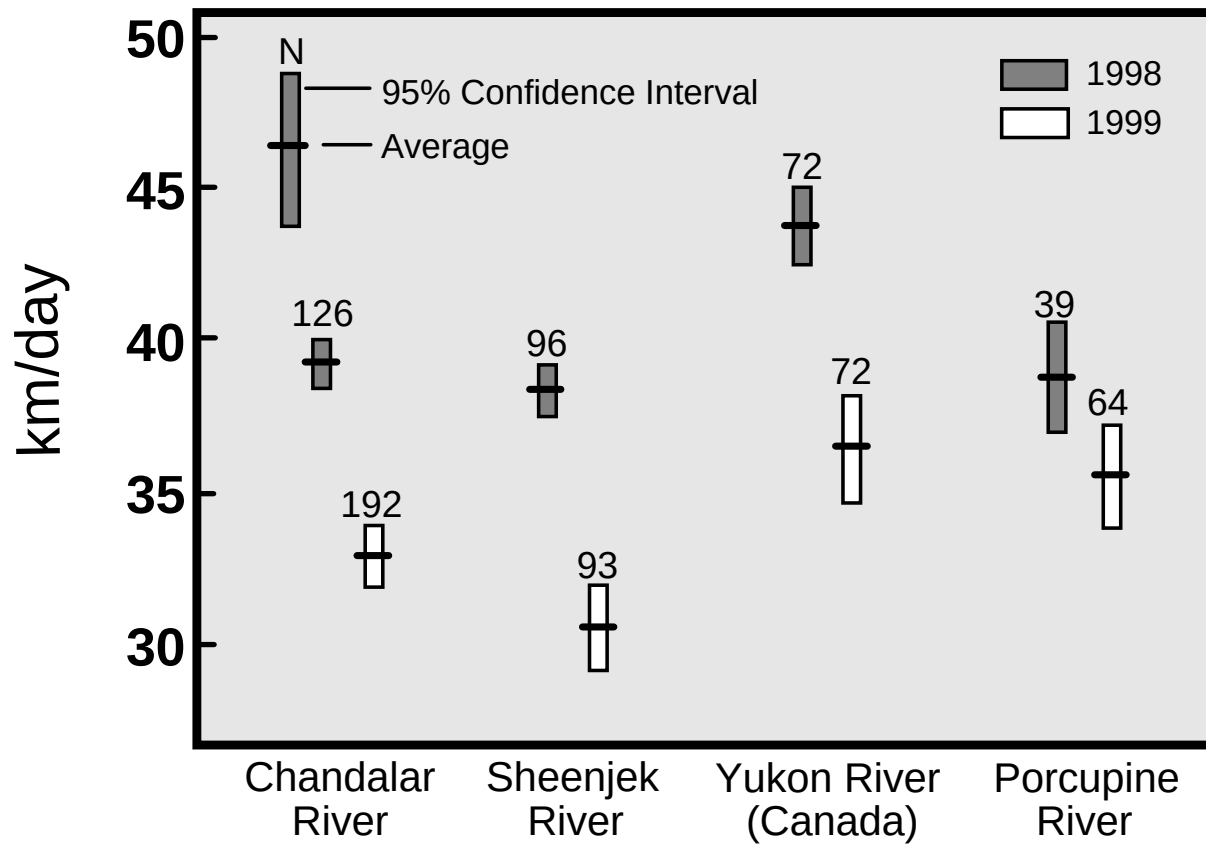


Figure 5. Movement rates (km/day) of radio-tagged chum salmon returning to the primary reaches of the upper Yukon River basin in 1998 and 1999.

Difference in fish movements were observed between the two years of the study. Average movement rates in 1998 (40.0 km/day) were significantly faster than in 1999 (33.5 km/day). This pattern was also observed for fish returning to the Chandalar, Sheenjek and Yukon Rivers in Canada, with fish from these areas traveling an average of 6.1 to 7.8 km/day faster in 1998. Although Porcupine River fish also tended to move upriver faster in 1998, the difference between the two years of the study was less distinct (Figure 5).

Movements between areas. Most fish returning upriver to spawn traveled to distinct reaches of the basin and remained there. However, in both years of the study, small numbers of fish exhibited movements between areas. These observations were limited to reaches in the general vicinity of the Yukon-Porcupine River confluence, and included fish moving from the Chandalar River to the Black and Sheenjek Rivers, from the Sheenjek River to the Chandalar and lower Porcupine Rivers, and from the lower and upper sections of the Porcupine River to the Sheenjek River. Fish were also recorded traveling past the Porcupine River border, moving back downstream to the lower reaches, and then returning to the upper Porcupine River. Some fish in the Chandalar and Sheenjek Rivers also exhibited repeated movements past the tracking stations on these tributaries. However, these stations were located in close proximity (i.e., 15-30 km) to reaches where spawning activity was observed, and may represent movements typical in localized spawning areas.

Upriver Movements by Gender.

Distribution of radio-tagged fish. Comparable numbers of male and female chum salmon were tracked upriver in both years of the study; 193 males and 169 females in 1998, and 315 males and 316 females in 1999 (Table 5). Differences in the distribution of males and females to the principle regions of the basin were not significant in 1998 ($P=.562$). Most fish of either sex returned to the Chandalar and Sheenjek Rivers. Substantial numbers were also tracked to upper reaches of the basin (distances greater than 600 km), while small numbers of fish remained in the Yukon Flats.

Table 5. Distribution of immediately released radio-tagged chum salmon to regions of the upper Yukon River basin by gender in 1998 and 1999. Percentages of the total are in parentheses.

Area	Minimum Distance	1998		1999	
		Male	Female	Male	Female
Yukon Flats ¹	11	7 (3.6)	10 (5.9)	68 (21.6)	94 (29.8)
Middle Reaches ²	400-500	122 (63.2)	101 (59.8)	162 (51.4)	124 (39.2)
Upper Reaches ³	> 600	64 (33.2)	58 (34.3)	85 (27.0)	98 (31.0)
Total		193	169	315	316

¹ Fish that moved past remote tracking stations located 11 km upriver from the tagging site but were not recorded by tracking stations further upriver.

² Chandalar and Sheenjek Rivers

³ Black River, Porcupine River and Yukon River in Canada.

The distribution exhibited by males and females was significantly different in 1999 ($P=0.012$). Most fish of both sexes returned to the middle reaches of the basin, however, a greater percentage of males (51.4%) were tracked to the Chandalar and Sheenjek Rivers compared to the female component of the sample (39.2%). Although increased numbers of both sexes remained in the Yukon Flats compared to 1998, the percentage of females (29.8%) was greater than the male component of the sample (21.6%). The proportion of fish tracked to upper reaches of the basin was comparable for both sexes; 27.0% of the males and 31.0% of the females.

Movement rates. Movement rates were comparable for male and female fish in 1998. Males moved an average of 40.4 km/day, ranging from 38.6 km/day for Sheenjek River fish to 44.6 km/day for fish traveling to the Yukon River in Canada. The fastest movement rate observed was 54.9 km/day for a male (56 cm in length) traveling past the Yukon River border. Movement rates for females

averaged 39.2 km/day, ranging from 36.7 km/day for Porcupine River fish to 44.5 km/day for fish traveling to the upper Yukon River. The fastest movement rate observed was 52.2 km/day for a female (60 cm in length) also traveling past the Yukon River border. Differences were observed between areas for both male and female fish. Females returning to the Yukon River in Canada traveled significantly faster than females in other reaches of the basin. Males exhibited a similar pattern, with Canadian Yukon River fish moving significantly faster than males returning to the Chandalar and Sheenjek Rivers, although the difference between Yukon River and Porcupine River males was not significant (Figure 6).

Movement rates exhibited by male and female fish in 1999 were also comparable. Males moved an average of 34.1 km/day, ranging from 31.1 km/day for Sheenjek River fish to 37.0 km/day for Porcupine River fish. The fastest movement rate observed was 49.8 km/day for a male (58 cm in length) traveling past the Yukon River border. Movement rates for females averaged 32.9 km/day, ranging from 29.4 km/day for Sheenjek River fish to 36.4 km/day for Yukon River fish in Canada. The fastest movement rate observed was 47.9 km/day for a female (62 cm in length) also traveling past the Yukon River border. Although differences between sexes were not significant, movement rates for males and females in the Chandalar and Porcupine Rivers were relatively distinct (Figure 6). Differences were observed between areas for both sexes. Males returning to the Chandalar, upper Yukon and Porcupine Rivers moved significantly faster than Sheenjek River males. Females traveling to the upper Yukon River moved significantly faster than Chandalar and Sheenjek females, however the difference with Porcupine River females was not significant (Figure 6).

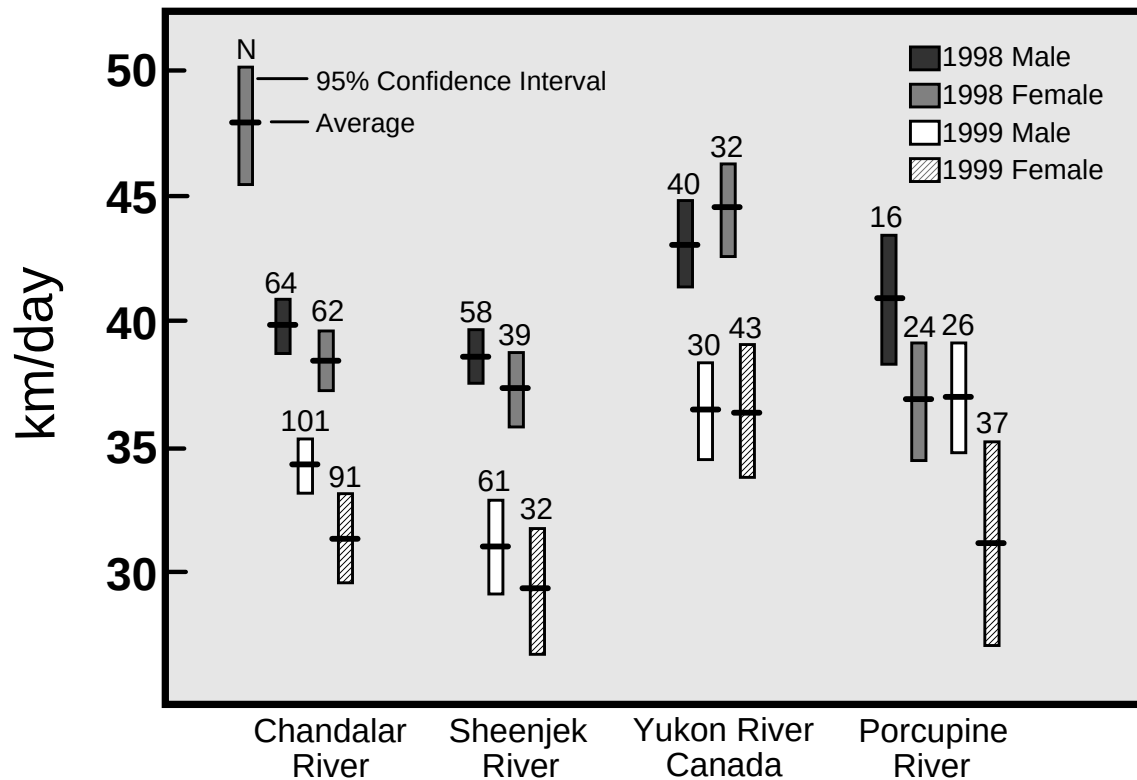


Figure 6. Movement rates (km/day) for male and female chum salmon radio-tagged in the upper Yukon River basin in 1998 and 1999.

Reductions in the average movement rates observed for males or females between 1998 and 1999 for all fish tracked to upper reaches of the basin (i.e., Chandalar River, Sheenjek River, Porcupine River and Yukon River in Canada) did not differ significantly ($p=.821$) between the sexes (Figure 7, left panel). Although the movement rates observed for males generally exceeded those for females among fish tracked to upper reaches of the basin, the differences were not uniform ($p=.035$) among the regional areas (Figure 7, middle panel). Average movement rates declined between 1998 and 1999 for fish traveling to upper reaches of the basin, but not uniformly among areas ($p=.001$). In

particular, Porcupine River fish (i.e., both sexes) exhibited a reduction in movement rates that was distinctly less than observed in fish from other areas (Figure 7, right panel).

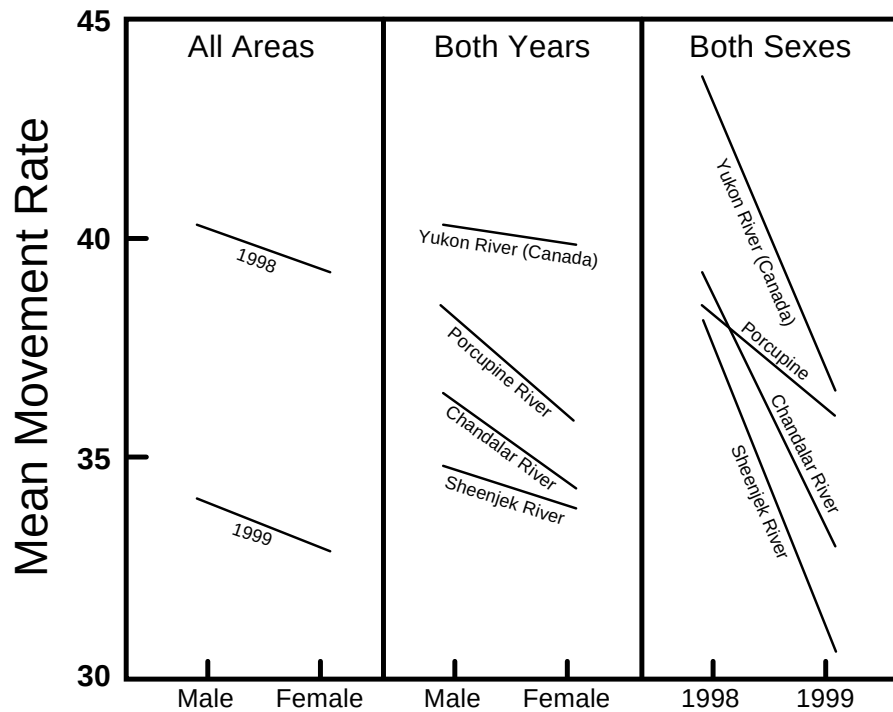


Figure 7. Comparisons of average movement rates (km/day) for radio-tagged chum salmon in the upper Yukon River basin based on analysis of variance of 1) males and females from all upriver areas by year, 2) males and females from both years of the study by area, and 3) all fish of both sexes in 1998 and 1999 by area.

Stock Composition

Stock composition estimates were derived for both years of the study, based on the distribution of radio-tagged fish and mark-recapture abundance estimates (Figure 8, Table 6). Although the entire run was sampled in 1999, tagging in 1998 was limited to the last four weeks of the run (Figure 9), representing approximately 87.5% of the return. Fish returning to the Chandalar River, Sheenjek River and Yukon River in Canada were the dominant stocks in 1998, comprising 34.2%, 27.0% and 21.0% of the return, respectively. The Porcupine River component, excluding Sheenjek River fish, was substantially less (13.4%), with fish traveling to the Black (2.5%), lower Porcupine (2.1%), Fishing Branch (3.6%) and upper Porcupine (5.2%) Rivers.

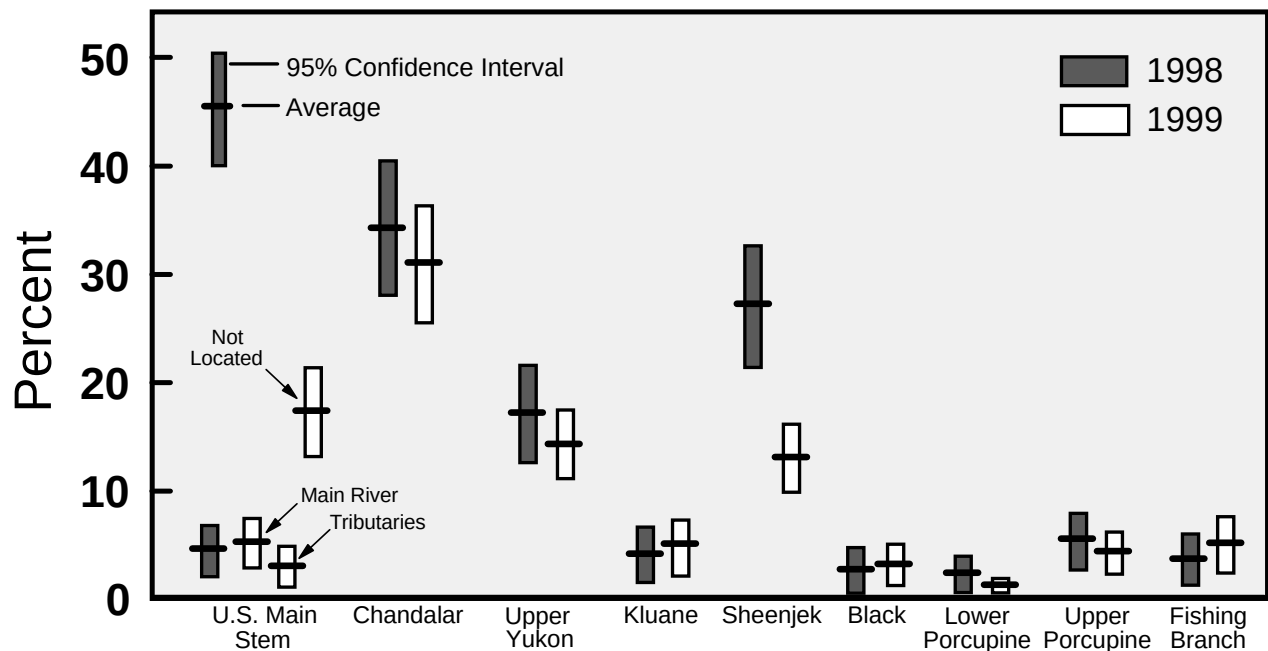


Figure 8. Stock composition estimates (%) of chum salmon in the upper Yukon River basin in 1998 and 1999. Proportions of the Yukon Flats component in main stem areas, minor tributaries or not located during fall aerial surveys in 1999, are indicated.

Table 6. Stock composition estimates (%) for chum salmon in the upper Yukon River basin in 1998 and 1999. Estimates are based on samples from the last four weeks of the run (August 23 to September 16) in 1998 and the entire run in 1999.

		1998	95% Confidence		1999	95% Confidence	
Drainage	Final Location	%	Lower	Upper	%	Lower	Upper
Yukon	Yukon Flats ¹						
	Tributaries ²				2.7	1.1	4.3
	Main stem ²				5.1	2.9	7.2
	Not relocated	4.4	2.1	6.8	17.2	13.2	21.3
	Chandalar River	34.2	28.0	40.3	30.9	25.7	36.2
	Upper Yukon ³	17.1	12.6	21.5	13.3	10.1	16.5
	Kluane River ³	3.9	1.6	6.3	4.8	2.2	7.3
Porcupine	Sheenjek River	27.0	21.4	32.5	13.0	9.9	16.0
	Black River	2.5	0.7	4.3	3.0	1.2	4.9
	Lower Porcupine	2.1	0.4	3.8	0.8	0.1	1.5
	Upper Porcupine ³	5.2	2.8	7.7	4.2	2.3	6.0
	Fishing Branch River ³	3.6	1.4	5.7	5.0	2.5	7.5

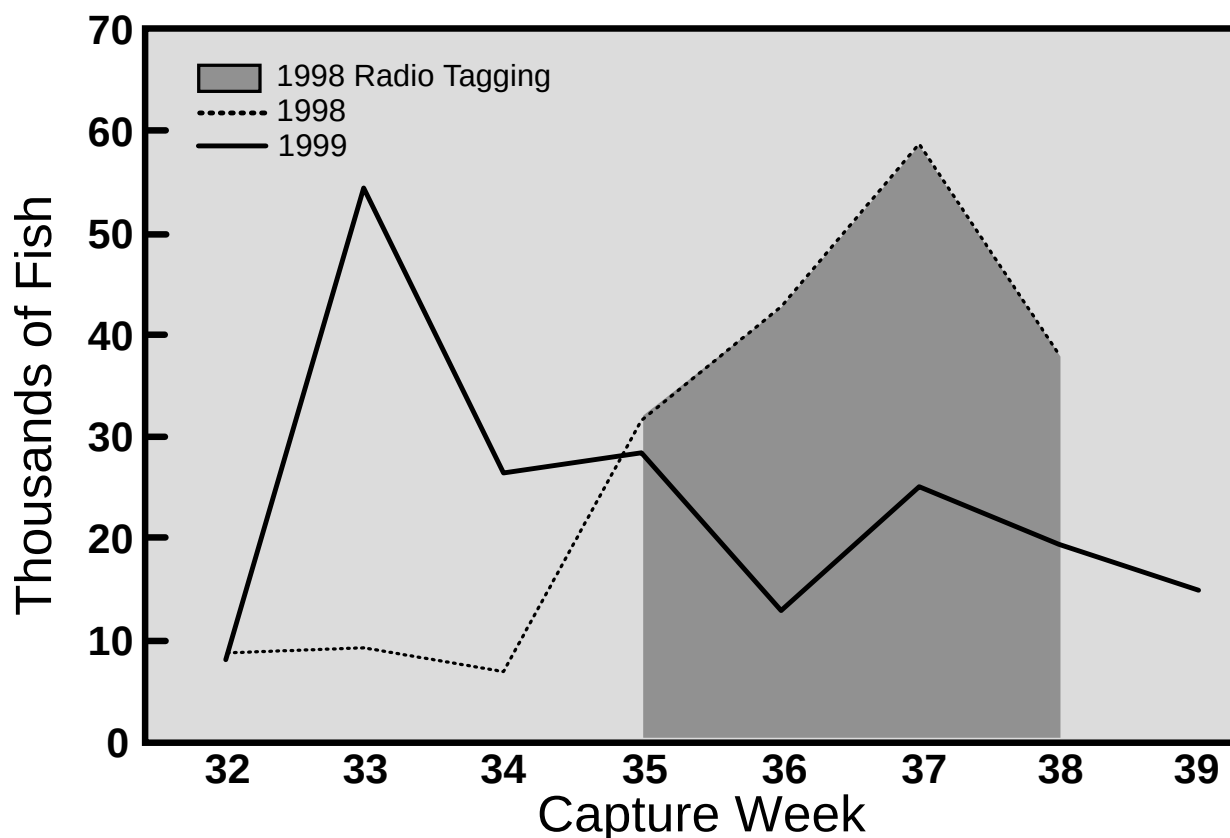
¹ Fish that moved past remote tracking stations located 11 km upriver from the tagging site but were not recorded by tracking stations further upriver.

² Fish located in secondary tributaries of the Yukon River main stem during aerial surveys conducted in 1999; comparable surveys were not flown in 1998.

³ Located in the Canadian section of the basin

Composition estimates in 1999 were similar for several stocks, including the Chandalar River (30.9%), Yukon River in Canada (18.1%), and Porcupine River component excluding the Sheenjek River (13.0%). However, estimates for the Sheenjek River (13.0%) were substantially less than the

previous year, while a greater proportion of fish were allocated to the Yukon Flats (25.0%); differences of 14% and 20.6%, respectively. The Yukon Flats component was comprised of fish located in minor tributaries (2.7%) and main stem areas (5.1%), as well as fish not found during fall

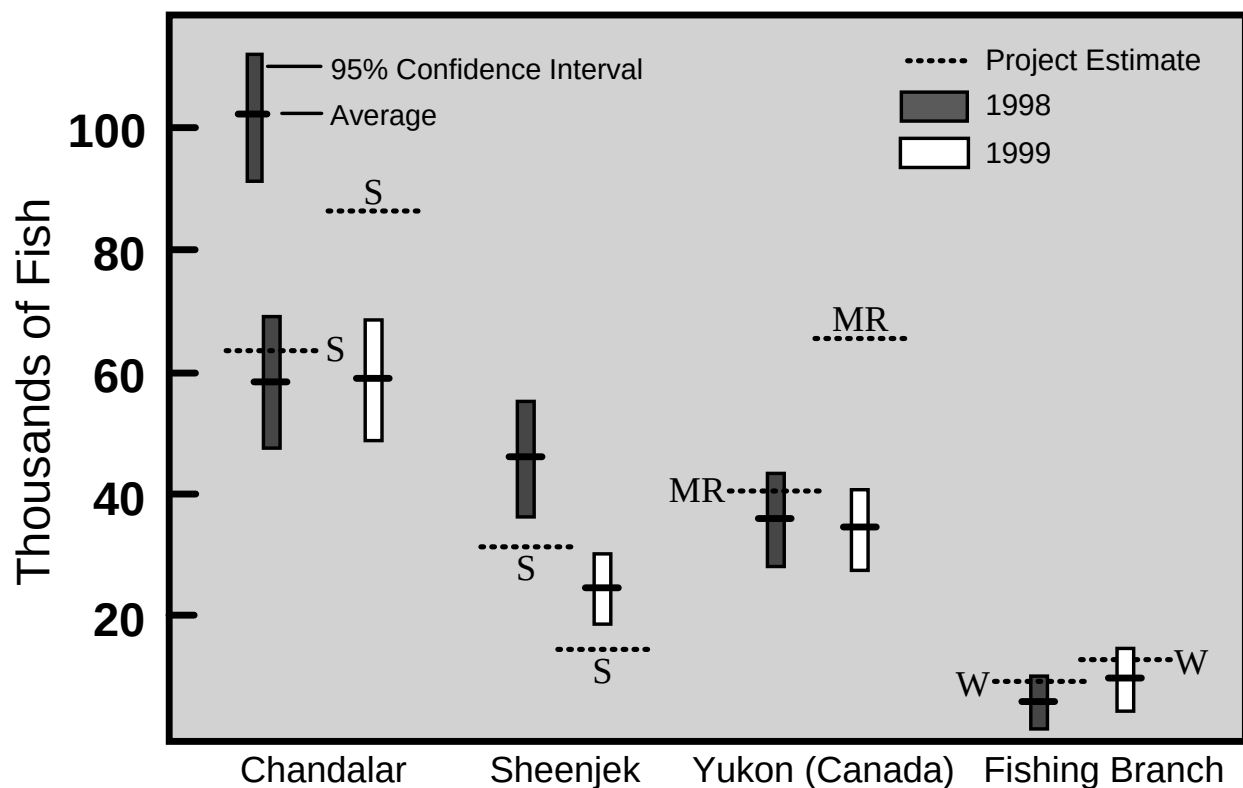


aerial surveys (17.2%).

Figure 9. Weekly run abundance of chum salmon in the upper Yukon River basin in 1998 and 1999. The shaded area represents the portion of the 1998 return sampled by radio tagging.

Stock abundance estimates, calculated using the composition and abundance estimates for the return, were compared with abundance information from other run assessment projects in the upper basin

including sonar counts from the Chandalar and Sheenjek Rivers, weir counts from the Fishing Branch River, and mark-recapture estimates for the Yukon River in Canada (Figure 10). Sonar counts from the Chandalar River were not significantly different than the stock abundance estimate in 1998, but significantly greater in 1999. Sonar counts from the Sheenjek River were significantly less than stock estimates in both 1998 and 1999, while mark-recapture estimates for the Yukon River in Canada were comparable in 1998 and significantly greater in 1999. Stock estimates for the Fishing Branch River were comparable to weir counts in both 1989 and 1999. Weir counts and abundance estimates from

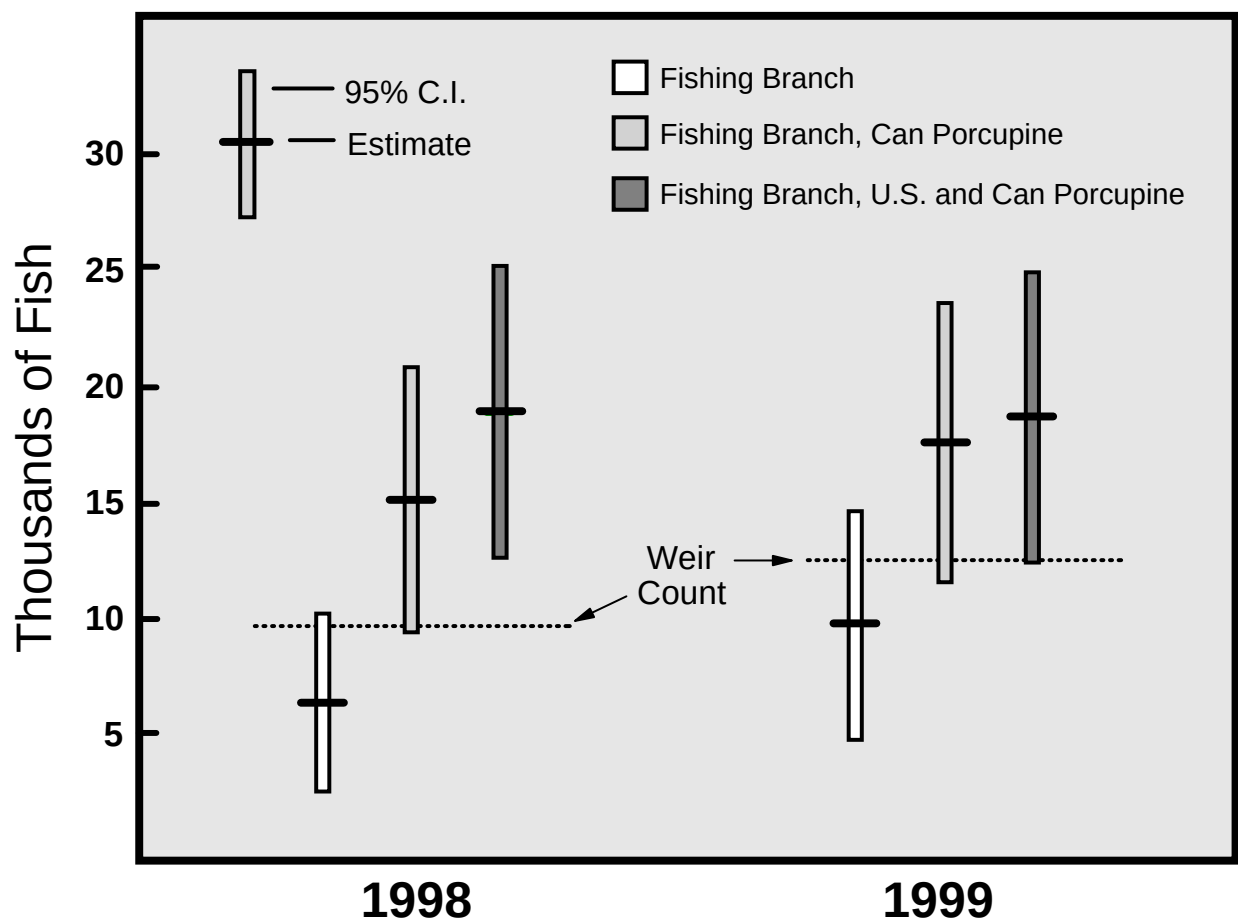


sonar and mark-recapture projects in 1998 were adjusted to reflect the numbers of fish present during the last four weeks of the run (Week 35-38) when fish were radio tagged.

Figure 10. Stock abundance estimates for Yukon River chum salmon in 1998 and 1999 compared with estimates from run assessment projects in the upper basin including sonar counts (S), weir counts (W), and mark-recapture (MR) estimates for the Yukon River in Canada. Project estimates in 1998 were adjusted to reflect the portion of the run sampled for radio tagging.

The most accurate abundance information in the upper basin is provided by counts at the Fishing Branch weir. To address concerns that fish, particularly those destined for upper reaches of the basin, were not reaching their final destination, we considered several scenarios of shortened migration (i.e., not completing movements to the intended spawning area) for radio-tagged fish traveling to the Fishing Branch River. The Fishing Branch stock was assumed to include 1) all fish tracked to the drainage (excluding Sheenjek and Black River fish), 2) all fish tracked to the Canadian portion of the drainage, and 3) only fish actually tracked to the Fishing Branch weir (Figure 11). The stock estimates derived for fish that traveled to the Fishing Branch in 1998 and 1999 were less than the actual weir counts, although within the 95% confidence interval. Estimates based on fish returning to the Canadian section of the drainage were greater than the actual weir counts, although still within the lower ranges of the 95% confidence interval. Estimates based on all fish returning to the drainage were significantly greater than the actual weir count in 1998, and substantially greater in 1999, although just within the lower 95% confidence interval.

Most (79.2%) of the fish in the Yukon River in Canada were located downstream from the Selkirk and Kluane tracking stations in 1998. A similar pattern was observed in 1999 when 72.9% of these fish traveled to this area, and comparatively few returned to the upper headwaters of the drainage. Combined composition estimates for fish located in Canadian index areas in the Yukon River main

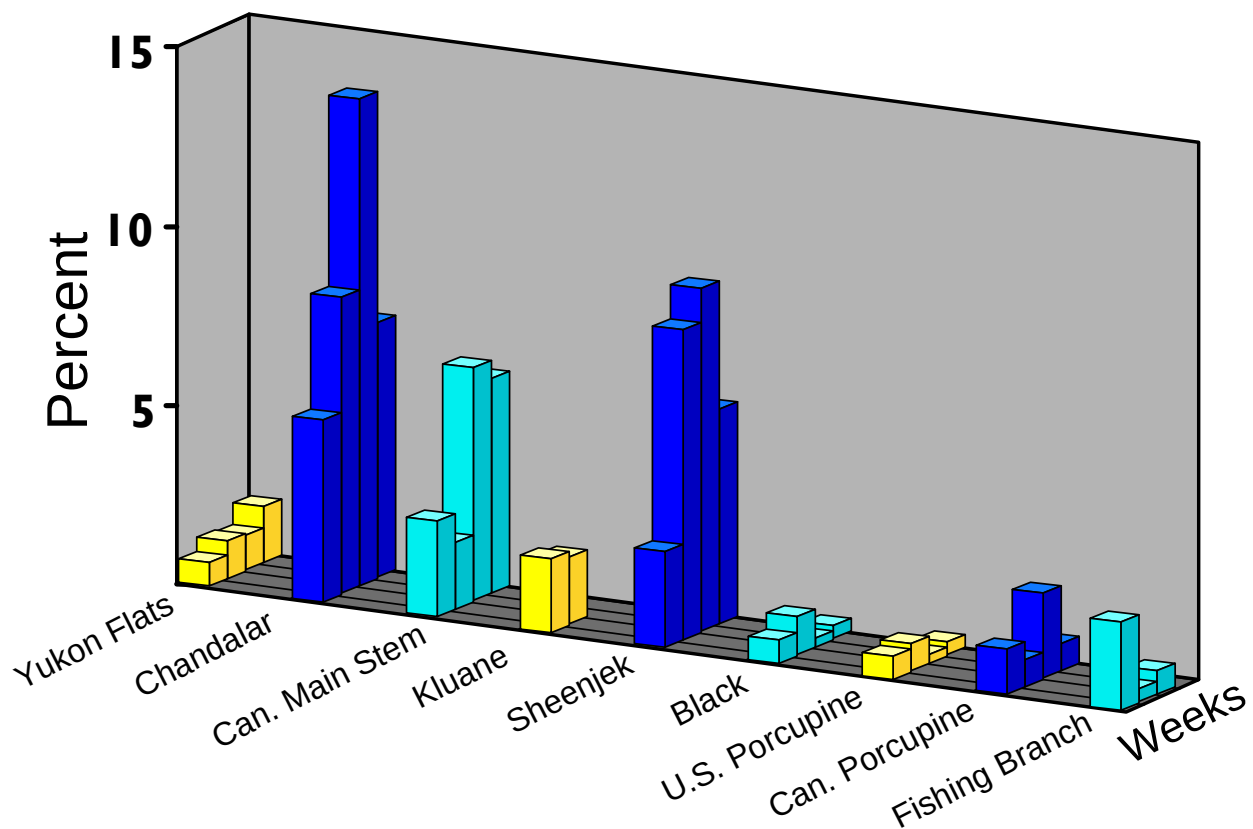


stem (between the Selkirk and Tatchun tracking stations) and Kluane River (between the lower and upper Kluane stations) ranged from 6.5% of the return sampled in 1998 to 6.9% of the entire return in 1999.

Figure 11. Comparison of escapement estimates of chum salmon returning to the Fishing Branch River in 1998 and 1999 if the radio-tagged sample included 1) only fish tracked to the Fishing Branch River, 2) fish tracked to the upper Porcupine and Fishing Branch Rivers, or 3) fish tracked to the lower Porcupine, upper Porcupine and Fishing Branch Rivers.

Stock composition estimates by week. Weekly stock composition estimates were developed for the last four weeks of the run (Week 34-38) in 1998 and the entire run (Week 32-39) in 1999 (Figure 9). Chandalar River fish were the dominant stock in 1998, ranging from 5.1% to 13.6% of the weekly returns (Figure 12). Sheenjek River fish were also abundant, particularly in Week 36-38 when they ranged from 6.0% to 9.6% of the weekly returns. The proportion of fish traveling to the Yukon River in Canada ranged from 3.8% to 6.5% of the return, however, the early component (Week 35-36) included substantial numbers of Kluane River fish, while the late run consisted only of upper Yukon River fish. Different timing patterns were observed for Porcupine River stocks, with Fishing Branch fish prevalent during the early component, compared to fish tracked to other reaches of the drainage. Relatively few fish remained in the Yukon Flats, ranging from 0.7% to 1.6% of the sampled return.

Figure 12. Weekly stock composition (%) for chum salmon returns in the upper Yukon River in 1998. Estimates are based on the distribution of radio-tagged fish, adjusted to account for the removal of tagged individuals by fisheries, and mark-recapture estimates for the last four weeks of the run (August 23 to September 19).



Chandalar River fish were again a dominant stock during weekly periods in 1999, particularly in Week 33 when they comprised 11.1% of the return (Figure 13). Sheenjek River fish were less abundant in 1999 compared to the previous year, comprising 0.4% to 2.6% of the run. Fish returning to the Yukon River in Canada were relatively abundant early in the run, particularly in Week 33 when

they comprised 5.0% of the return. The magnitude of this component in Week 33 was due largely to the presence of Kluane River fish which made up 3.3% of the return. Few fish traveled to the Kluane River later in the run, with most fish returning to other sections in the upper drainage (i.e., upper Yukon River fish). Most chum salmon returning to the Porcupine River in Canada were early-run fish, particularly those returning to the Fishing Branch River, which composed 3.3% of the return in Week 33. Upper Porcupine River fish exhibited a more normal distribution, and were present throughout the run, ranging from 0.2% to 1.0% of the return. Lower Porcupine and Black River fish were present in small numbers throughout the run, although Black River fish were most abundant (1.5%) during Week 33. Fish located in minor tributary streams in the Yukon Flats were the primary stock present during the first week of the run (Week 32), comprising 1.5% of the return. Although tributary fish were most prevalent during the early run, small numbers were present during the middle and late run as well. Fish were also located in main stem areas, ranging from 0.2% to 1.1% of the return. Substantial numbers of fish were tracked to the Yukon Flats, but not located during aerial surveys of tributaries and main stem areas. These fish ranged from 0.8% to 3.4% of the return, and were most abundant during the early (Week 33) and late (Week 37-39) run.

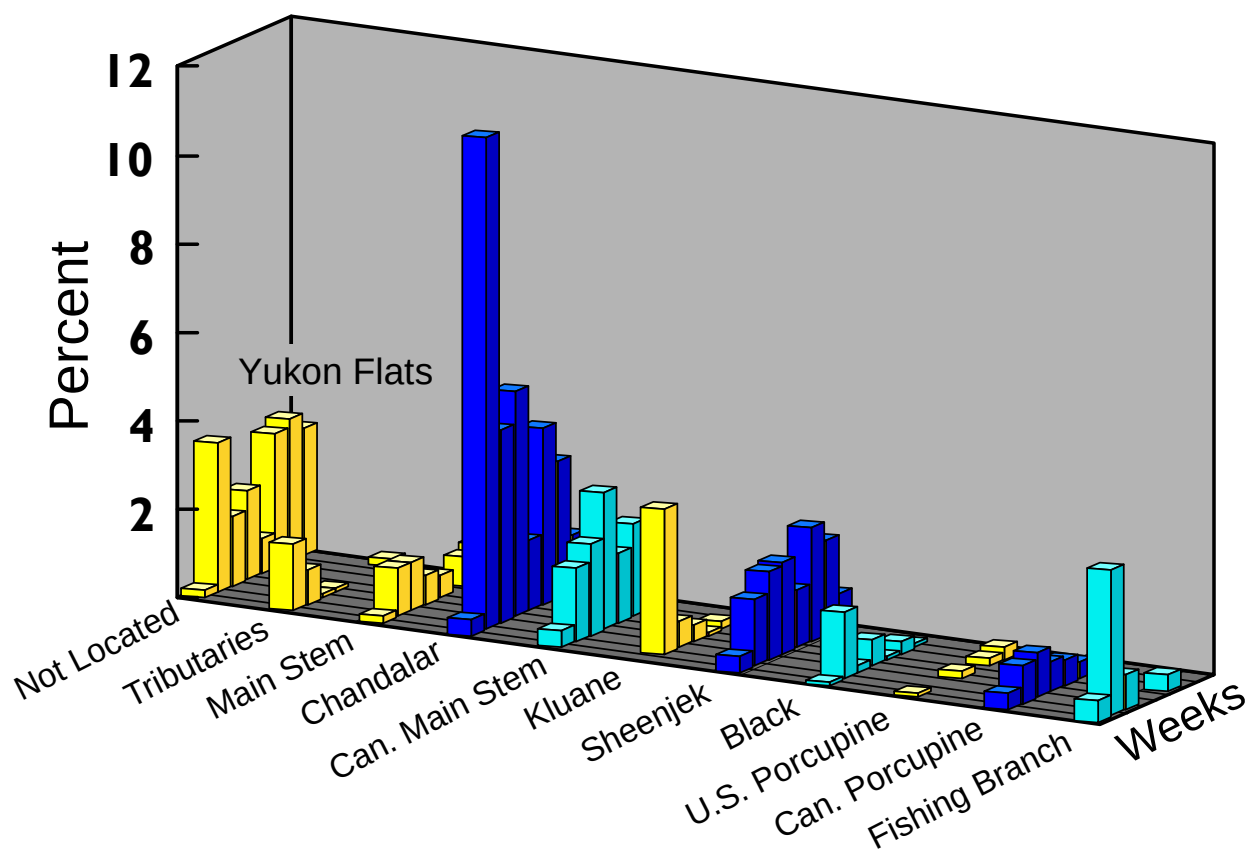


Figure 13. Weekly stock composition (%) for chum salmon returns in the upper Yukon River in 1999. Estimates are based on the distribution of radio-tagged fish, adjusted to account for the removal of tagged individuals by fisheries, and mark-recapture estimates for the entire run (August 4 to September 22).

Tagged-Untagged Ratios

Tagged-untagged ratios were compared for fish recovered in the U.S. and Canadian fisheries, and at the Fishing Branch weir in 1998 and 1999. Samples were grouped into lower (from the tagging site to Circle, distances ≤ 600 km) and upper (upriver from Circle and the upper Porcupine River, distances > 600 km) reaches of the basin. Twenty-six fish of 22,375 fish recovered in 1998 were radio tagged; seven fish of 4344 fish (.0016%) in lower reaches, and 19 fish of 18,031 fish (.0011%) in upper reaches. Difference in tagged-untagged ratios for fish below and above Circle were not

significant ($P = 0.459$). In 1999, 84 fish of 59,432 fish recovered were radio tagged, including 21 fish of 16,784 fish (.0013) in lower reaches and 63 fish of 42,648 fish (.0015) in upper reaches. Difference in tagged-untagged ratios for fish caught below and above Circle were not significant ($P = 0.30$).

Holding Experiments

Resumption of upriver movements. Of the 50 fish radio tagged in 1996 (Table 2), 47 fish resumed upriver movements, including 24 (96.0%) fish tagged and released immediately after capture and 23 (92.0%) fish held under crowded conditions in the fish wheel live box prior to tagging and release. Forty-one of these fish moved upriver without being recaptured by the tagging site fish wheels, including 23 immediately released fish and 18 held fish (Table 7), and were included in the analysis. Immediately released fish averaged .79 days to move past the Rapids tracking stations, ranging from .21 days to 3.08 days. Held fish averaged 2.02 day to move past the stations, ranging from .38 days to 6.17 days.

Similar results were observed in 1998 when 227 (94.6%) fish tagged as part of the experiment resumed upriver movement, including 76 (95.0%) immediately released fish tagged with radio and spaghetti tags, 77 (96.5%) immediately released fish tagged with only radio tags, and 74 (92.5%) fish tagged with radio and spaghetti tags after being held under crowded conditions in the fish wheel live box. Most of these fish moved upriver without being recaptured by the tagging site fish wheels, including 129 immediately released fish (60 fish with radio and spaghetti tags, and 69 fish with only radio tags), and 63 held fish (Table 7). Immediately released fish averaged .65 days (radio and spaghetti tag) and .51 days (radio tag only) to moved past the Rapids tracking stations, ranging from

.25 days to 2.17 days and .25 days to 4.17 days, respectively. Held fish took an average of 2.05 days to pass the stations, ranging from .25 days to 9.88 days.

Table 7. Elapsed time from release for radio-tagged chum salmon to move past remote tracking stations located 11 km upriver from the tagging site in 1996, 1998 and 1999, by handling method. Fish were tagged and released immediately after capture, or held in fish wheel live boxes for 4-6 hours under crowded or uncrowded conditions prior to tagging and release.

Year	Treatment	Tag Type		N	Average (Days)	SD	Range	
		Radio	Spaghetti				Lower	Upper
1996	Immediate release	x	x	23	0.79	0.2	0.21	3.08
	Held, crowded	x	x	18	2.02	0.4	0.38	6.17
1998	Immediate release	x	x	60	0.65	0	0.21	4.17
	Immediate release	x		69	0.51	0.1	0.25	2.17
	Held, crowded	x	x	63	2.05	0.2	0.25	9.88
1999	Immediate release	x	x	176	0.99	0.1	0.25	5.96
	Held, crowded	x	x	33	2.85	0.4	0.5	8.67
	Held, uncrowded	x	x	31	2.52	0.3	0.38	13.67

In 1999, 319 (92.3%) fish tagged as part of the experiment resumed upriver movements, including 230 (93.5%) immediately released fish, 44 (88.0 %) fish held under crowded conditions, and 45 (90.0%) fish held under uncrowded conditions. Differences in the proportion of fish that moved upriver among the three treatment groups were not significant ($P=.310$). A total of 240 fish moved upriver without recapture by the tagging site fish wheels, including 176 immediately released fish,

33 held/crowded fish, and 31 held/uncrowded fish (Table 7). Immediately released fish averaged .99 days to move past the Rapids tracking stations, ranging from .25 days to 5.96 days. Fish held under crowded conditions took an average of 2.85 days to pass the stations with a range of .5 days to 8.67 days, while fish held under uncrowded conditions averaged 2.52 days with a range of .38 days to 13.67 days.

Consistent patterns were observed for fish in the treatment groups. Travel times exhibited by immediately released fish were comparable during the three years of the study based on 95% confidence intervals (Figure 14). Differences observed in 1998 between fish tagged with both radio and spaghetti tags and those with only radio tags were not significant. Travel times were also comparable for held fish during the three years of the study, although the average times observed in 1999 were higher than in the previous years. Differences observed in 1999 between fish held under crowded and uncrowded conditions were not significant. However, travel times of held fish were significantly greater than those of immediately released fish during all three years of the study (Figure 14). Travel times of held fish were also more variable with ranges of 5.8 days (.38 days to 6.17 days) in 1996, 9.6 days (.25 days to 9.88 days) in 1998, and 13.3 days (.38 days to 13.67 days) in 1999, compared to 2.87 days (.21 days to 3.08 days) in 1996, 1.92 days (.25 days to 2.17 days) in 1998 and 5.71 days (.25 days to 5.95 days) in 1999 for immediately released fish (Table 7).

Distribution of radio-tagged fish. A total of 180 radio-tagged fish moved upriver in 1998 without capture in mid-river fisheries or research fish wheels. This sample included 126 immediately released fish (59 fish tagged with radio and spaghetti tags and 67 fish tagged with only radio tags) and 54 fish held under crowded conditions (Table 8). Differences in distribution between the two

groups of immediately released fish where not significant ($P=.975$), suggesting that spaghetti tagging had negligible effect on fish performance. Comparable proportions of held fish and immediately released fish were tracked to middle reaches of the upper basin, specifically the Chandalar and Sheenjek Rivers. However, a greater proportion of held fish remained in the Yukon Flats, while a greater proportion of immediately released fish were tracked to upper reaches of the basin. Differences in distribution of held and immediately released fish were significant ($P< .05$).

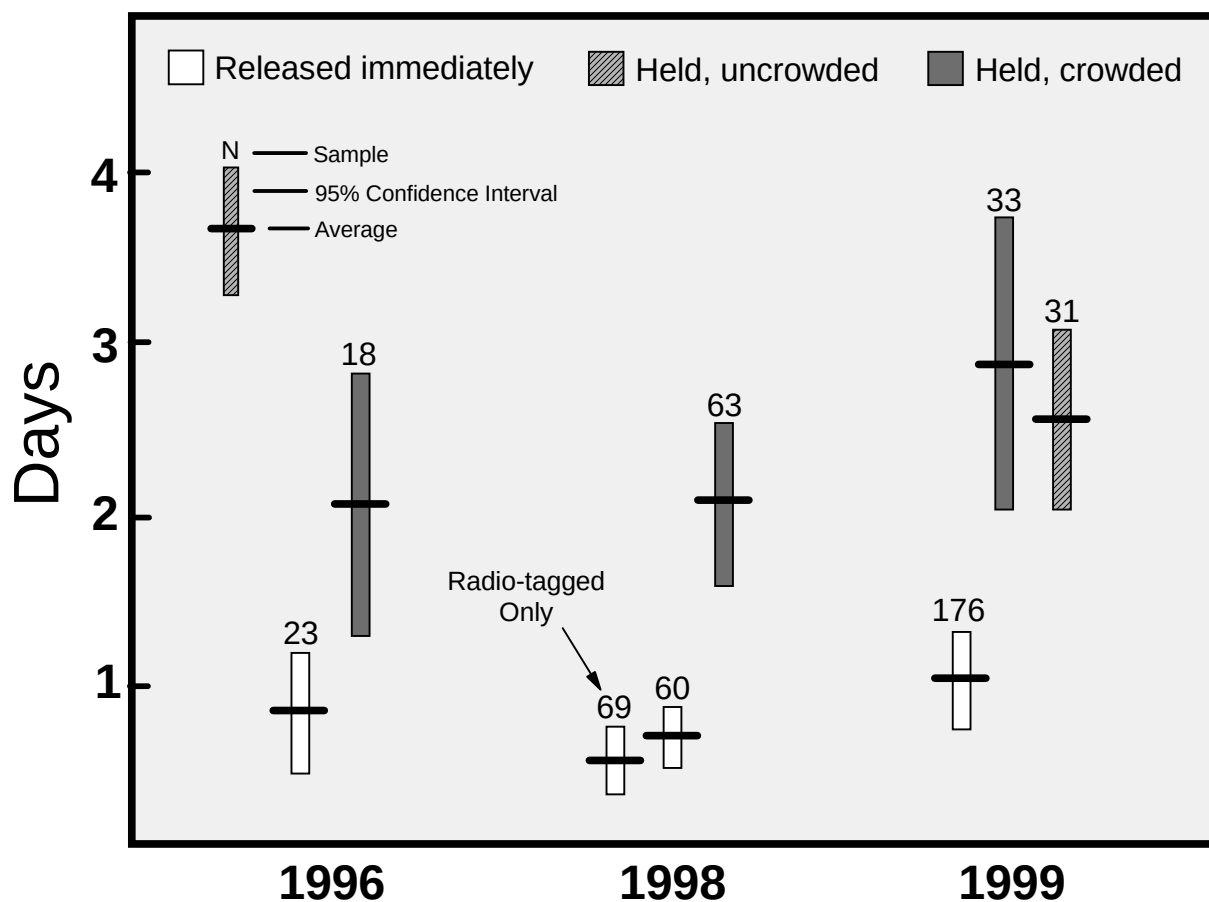


Figure 14. Comparison of elapsed times from release for radio-tagged chum salmon to move past remote tracking stations located 11 km upriver from the tagging site in 1996, 1998 and 1999 by handling method. Fish were tagged and released immediately after capture, or held under crowded and uncrowded conditions in fish wheel live boxes for 4-6 hours prior to tagging and release.

Table 8. Distribution of radio-tagged chum salmon recorded past tracking stations in the upper Yukon River basin during 1998 by handling method. Fish were tagged with both radio and spaghetti tags unless otherwise specified. Minimum distances were calculated as the distance (km) from the Rapids tracking stations to tracking stations at upriver sites that separated sections of the drainage. Percentage of treatment sample is in parentheses.

Area	Minimum Distance (km)	Treatment		
		Immediate Release	Radio Tags Only Immediate Release	Held Crowded
Yukon Flats ¹	11	1 (1.7)	2 (3.0)	13 (24.1)
Middle Reaches ²	400-500	37 (62.7)	42 (62.7)	32 (59.2)
Upper Reaches ³	> 600	21 (35.6)	23 (34.3)	9 (16.7)
Total		59	67	54

¹ Fish that moved past remote tracking stations located 11 km upriver from the tagging site but were not recorded by tracking stations further upriver.

² Chandalar and Sheenjek Rivers

³ Black River, Porcupine River and Yukon River in Canada

Similar distribution patterns were observed in 1999. A total of 197 fish moved upriver without capture in mid-river fisheries or research fish wheels, including 146 immediately released fish, 26 fish held under crowded conditions, and 25 fish held under uncrowded conditions (Table 9). Differences in distribution between the two groups of held fish were not significant ($P=.393$), suggesting that fish held under uncrowded and crowded conditions had a similar response. A greater percentage of immediately released fish remained in the Yukon Flats than observed in 1998, however, the percentage of held fish was also substantially greater. A greater proportion of immediately released fish were observed in upriver areas which was similar to observations in 1998. Differences

in the distribution of held fish (crowded and uncrowded treatments combined) and immediately released fish were significant ($p < .001$).

Table 9. Distribution of radio-tagged chum salmon recorded past tracking stations in the upper Yukon River basin during 1999 by handling method. Minimum distance was calculated as the distance (km) from the Rapids tracking stations to upriver tracking stations that separated regions of the drainage. Percentage of treatment sample is in parentheses.

Area	Minimum Distance (km)	Treatment		
		Immediate Release	Held Crowded	Held Uncrowded
Yukon Flats ¹	11	28 (19.2)	16 (61.5)	13 (52.0)
Middle Reaches ²	400-500	69 (47.2)	8 (30.8)	6 (24.0)
Upper Reaches ³	> 600	49 (33.6)	2 (7.7)	6 (24.0)
Total		146	26	25

¹ Fish that moved past remote tracking stations located 11 km upriver from the tagging site but were not recorded by tracking stations further upriver.

² Chandalar and Sheenjek Rivers

³ Black River, Porcupine River and Yukon River in Canada

A similar pattern was also observed in the distribution of immediately released fish, tagged throughout the entire study, that moved upriver without being recaptured compared to those that were caught and released at the Rampart fish wheels. The sample included 631 fish that were not recaptured and 85 fish recaptured fish in 1999 (Table 10). The percentages of the two groups tracked to the Chandalar and Sheenjek Rivers were comparable. However, a greater percentage of fish recaptured at Rampart remained in the Yukon Flats, while a greater percentage of non-recaptured fish

were tracked to upper reaches of the drainage. Differences in the distribution between the two groups were significant ($p < .001$). Only nine fish were reported recaptured in the Rampart fish wheels in 1998 making similar comparisons of questionable merit.

Table 10. Chum salmon distribution in 1999 comparing fish that moved upriver without recapture in fish wheels and those recaptured and released in recovery fish wheels at the village of Rampart. All fish were tagged with radio and spaghetti tags immediately after capture and released. Minimum distance was calculated as the distance (km) from the Rapids tracking stations to upriver tracking stations that separated different regions of the drainage. Percentage of treatment sample is in parentheses.

Area	Minimum Distance	Not Recaptured	Recaptured in Rampart Fish Wheels
Yukon Flats ¹	--	162 (25.7)	39 (45.9)
Middle Reaches ²	400-500	286 (45.3)	33 (38.8)
Upper Reaches ³	> 600	183 (29.0)	13 (15.3)
Total		631	85

¹ Fish that moved past remote tracking stations located 11 km upriver from the tagging site but were not recorded by tracking stations further upriver.

² Chandalar and Sheenjek Rivers

³ Black River, Porcupine River and Yukon River in Canada

Periodically fish tracked upriver were later detected moving downriver past the Rapids tracking station, including fish that had remained in the Yukon Flats as well as upriver sections of the drainage. This group could conceivably consist of fish that had spawned and were drifting downriver, were injured in upriver fisheries, had been adversely affected by the capture and handling

procedures, or were destined for other reaches of the drainage (as described previously). Most of the fish that remained in the Yukon Flat and exhibited downriver movement had been recaptured in the Rampart fish wheels; 7 (77.8%) in 1998 and 16 (76.2%) fish in 1999, suggesting that at least a portion of the recaptured fish had been adversely affected by handling.

Discussion

Run Characteristics

Salmon returning to the Yukon River basin support important commercial and subsistence fisheries within the region. These returns are a major source of food in many remote communities, and often constitute the primary source of income. Management of the resource is complicated by recent declines in salmon returns and the international nature of the drainage, which makes it necessary to address harvest allocation issues in both the U.S. and Canada. Detailed information on run characteristics, including run abundance, migratory patterns, stock composition and timing, is needed to improve management of fisheries and facilitate conservation efforts. However, determining run characteristics in drainages like the Yukon River basin is a challenging task. The immense size, remoteness, and turbid nature of many reaches makes needed information difficult to obtain. In addition to harvest records from fisheries along the Yukon River main stem, fall chum assessment programs have been conducted over a number of years to provide abundance estimates for selected sections of the upper Yukon River basin. These programs have focused on stocks thought to comprise the primary components of the run, including Chandalar and Sheenjek River fish in the U.S., and fish returning to the Fishing Branch River and the Yukon River in Canada. The joint mark-recapture and radio tagging program initiated in 1996 was an attempt to provide information that encompassed the entire run.

Chum salmon returning to the Chandalar and Sheenjek Rivers were important components during both years of our study, comprising between 61.2% of the run sampled in 1998, and 43.9% of the entire return in 1999. However, differences between years were observed for these two stocks. Chandalar River fish were the dominant stock during both years of the study, ranging from 34.2% to 30.9% of the return. In addition, composition estimates indicate that fish returning to this area mirrored weekly abundance estimates in both 1998 (i.e., during the period sampled) and 1999, suggesting that this stock was a primary component throughout the return. This pattern was observed to a lesser extent for Sheenjek River fish in 1998 with the exception of the first week of tagging (Week 35), suggesting that, although a dominant component of the run (27.0%), the timing of this stock was slightly later than observed for the Chandalar River. Sheenjek River fish comprised a significantly lower proportion of the return in 1999 (13.0%). Sonar counts for the Sheenjek River also indicated a substantial decline in escapement in 1999 (Yukon River Joint Technical Committee 2000). Sheenjek River fish were present throughout the run, and with the exception of the peak (Week 33), also tended to reflect the general timing of the return.

Fishing Branch River fish constituted a relatively minor component of the return, ranging from 3.6% in 1998 to 5.0% in 1999. The 1998 estimate is undoubtedly lower than the actual proportion for this stock since we only sampled the last four weeks of the run in 1998, and timing data suggests that these fish move through the basin early in the return. However, chum abundance in 1998 during the early run was relatively low (Figure 9), and it is unlikely that the composition estimate would change substantially since other stocks also appear to be present during this period. Weir counts of chum salmon returning to the Fishing Branch River have declined severely in recent years, ranging from a high of 353,282 fish in 1975 to a low of 12,904 fish in 1999 (Yukon River Joint Technical

Committee 2000). A comparison with abundance estimates from other assessment projects in the upper basin (Yukon River Joint Technical Committee 2000) suggests that Fishing Branch escapements were between 7.8% and 6.6% of the return during the two years of the study; levels comparable to our findings.

Stocks returning to the Yukon River in Canada, including upper Yukon and Kluane River fish, comprised a substantial component of the run, ranging from 18.1% to 21.0% of the return. Kluane River fish were less abundant and primarily an early run stock, although fish returning to this area were also present during later periods. This was particularly evident in 1999 when the entire run was sampled, with Kluane River fish comprising 4.3% of the return during the first half of the run compared to 0.3% during the last half of the run. Upper Yukon River fish peaked near the middle of the return, although this stock was also more prevalent later in the run. The spatial distribution of this stock was similar during both years of the study, with 68.1% and 64.4% of the fish in the Canadian section of the Yukon River remaining downriver of the Selkirk tracking station compared to 18.1% and 20.5% tracked upriver of this site (Table 11). Comprehensive data on where these fish spawned is not available, although ancillary information from aerial surveys and assessment projects indicated that fish spawned in both secondary tributaries (e.g., radio-tagged fish were observed at the Chandindu River weir downstream of Dawson) and main stem areas. Milligan et al. (1986) reported that Yukon River chum salmon radio tagged above the U.S.-Canada border in 1982 were tracked to the Klondike, White, Donjek, Kluane, and Teslin Rivers as well as sections of the Yukon River main stem. Whether fish in the White River were actively spawning was not determined due to the turbid nature of this area. Although the method used to locate radio-tagged fish differed during the 1982 study (i.e., aerial tracking versus remote tracking stations), the distribution data has been summarized

to be comparable to information from the present study (Table 11). Although the proportions of fish tracked to the three areas within the Canadian section varied among years, the differences observed were not significant ($p=.258$).

Table 11. Distribution of radio-tagged fall chum salmon in the Canadian section of the Yukon River in 1982 (Milligan et al. 1986) and 1998-1999 (this study). Percentages of the total numbers tracked to this region are in parentheses.

Area	1982	1998	1999
Downriver of Selkirk and Kluane tracking stations	34 (50.0)	49 (68.1)	47 (64.4)
Upriver of Selkirk tracking station	20 (29.4)	13 (18.1)	15 (20.5)
Upriver of Kluane tracking station	14 (20.6)	10 (13.9)	11 (15.1)
Total	68	72	73

Relatively little information is available on minor stocks of chum salmon in the upper basin. During this study, small numbers of fish returned to the Black River, and reaches of the lower and upper Porcupine River. These stocks were present throughout the run, although Black River fish in 1999 were most abundant early (Week33). Fish in the upper Porcupine River were more abundant than the lower reaches of the drainage, and were more prevalent during the first half (3.0% of the return) than the last half of the run (1.2% of the return).

Fish Status in the Yukon Flats

Most fish in 1998 returned to terminal reaches of the basin, often to known spawning areas. The numbers of fish in non-terminal areas, such as the Yukon Flats and lower Porcupine River were

relatively small. However, the non-terminal component in 1999 was substantially larger, which raises questions regarding the stock composition estimates obtained. Central to these questions is the status of fish tracked to the Yukon Flats. The Yukon Flats is a relatively unique section of the basin, comprised of an extensive and complex series of side channels, sloughs and minor tributaries flowing into the main river. Determining the status of fish in this area is difficult. Main stem reaches are extremely turbid, making visual observations impossible, and the immense size and remote nature of the area limits regular access. The flat topography and braided nature of this section of the river makes monitoring radio-tagged fish with tracking stations difficult. For this reason, tracking stations were deployed in a manner that cordoned off major tributaries within the area and sections of the main stem further upriver.

Underlying the stock composition estimates, based on the distribution of radio-tagged fish, is the presumption that fish tracked to upriver reaches were spawning in these areas. This assumption is well founded for fish traveling to terminal reaches. However, alternatives exist for fish tracked to non-terminal reaches. In addition to returning to spawning areas in main stem reaches or associated tributaries, fish remaining in the Yukon Flats could have been caught in local fisheries but not reported, or died while in transit to upriver reaches due to predation, handling at the Rapids tagging site, or encounters with fishing gear. Relatively few fish remained in the Yukon Flats during 1998, minimizing potential error to the composition estimates. These fish were observed throughout the period sampled, although they were somewhat more prevalent later in the run. However, the number of fish remaining in this area in 1999 was significantly greater than observed the previous year, representing a significant potential bias for stock composition estimates.

Efforts were made to address the issue of Yukon Flats fish in 1999. Aerial surveys were not flown routinely during the study due to insufficient personnel and funds. This approach provided acceptable information in 1998 when most fish traveled to terminal reaches of the drainage – areas likely used for spawning. Due to the large numbers of radio-tagged fish tracked to the Yukon Flats in 1999, several surveys were conducted in late fall to determine the status of fish in this area. The data collected during these surveys were limited. Although the timing of the flights ensured that fish were not actively moving, most of the fish located were dead, and may have drifted downstream from areas further upriver. Due to the size and complex nature of the area, not all main stem reaches and associated tributaries could be surveyed. The areas surveyed included most of the main river, certain tributaries thought to constitute likely spawning areas, and side channels and sloughs in certain sections of the main stem. Flights were also limited in duration due to winter conditions and limited daylight, so that the surveys were somewhat cursory in nature (e.g., not all sections of a selected tributary or side channel complex could be surveyed) and focused on areas that were readily accessible.

The fish detected in main stem tributaries in the Yukon Flats were located in areas ranging from upper headwaters to reaches near the confluence with the Yukon River, and were thought to represent spawning fish. Although tributary fish were scattered throughout the run, the greatest proportion was observed during the first week of tagging (Week 32) when this component comprised a relatively large percentage of the return. Tributary fish in Week 32 were possibly late-run summer chum salmon, although summer chum salmon typically do not return to areas upriver from the Rapids. Tissue samples were taken from radio-tagged fish in 1999 to provide genetic information. These samples are

being analyzed by the USFWS, and when completed will likely address this issue, since methods to distinguish between summer and fall chum salmon are well established (Wilmot et al. 1992).

Fish were also located in main stem areas of the Yukon Flats. Although their status was not determined, a portion or all of these fish may have spawned in this area. Chum salmon are known to spawn in side channels and sloughs of main rivers, often in areas with upwelling groundwater (Milligan et al. 1986, Eiler et al. 1992, Helle and Hoffman 1995). Chum salmon spawn in main stem areas in the upper Yukon River, specifically in reaches between the Selkirk and Tatchun tracking stations where side channels and sloughs clear periodically during the fall, making direct observations possible (Milligan et al. 1986). In addition to salmon, other fish species have been observed in spawning condition in main stem areas of the Yukon Flats including inconnu (*Stenodus leucichthys*), humpback whitefish (*Coregonus pidscian*) and Bering cisco (*C. lauretus*) (Brown 2000). Some of the radio-tagged chum salmon tracked during fall surveys were located immediately downriver from main stem tributaries and may represent tributary fish that drifted downriver after spawning.

Another possibility is that radio-tagged fish died prematurely as they moved upriver, i.e. fish remaining in the Yukon Flats were actually destined for reaches further upriver, but were unable to complete their spawning migration due to injuries or stress resulting from capture and handling. A primary assumption in tagging studies is that capture and handling procedures do not adversely affect the fish sampled (i.e., tagged and untagged fish behave the same), or that any affect is limited in severity and duration, and ultimately has negligible impact. For example, a common response observed in tagging studies is for fish to hold in a localized area or drift with the current immediately after release prior to resuming normal movement patterns. Extreme care was taken during the present

study to prevent physical injury and minimize stress which can potentially affect performance and reduce survival. Numerous studies have been conducted to assess tagging effects on fish, and anomalous behavior has periodically been reported (McCleave and Stred 1975, Mellas and Haynes 1985, Pella et al. 1988, Haegele 1990, Matthews and Reavis 1990, Brown and Eiler 2000). However, results can vary substantially depending on a number of factors including the handling methods used, environmental variables, physiological condition of the fish, and species. In addition to immediate impact, some studies have also reported latent effects from handling (Kelsch and Shields 1996).

Underwood et al. (2002) reported that marked-unmarked ratios of Yukon River chum salmon, tagged at the Rapids with external spaghetti tags and recovered in upriver fisheries, declined significantly the farther upriver the fish traveled, and speculated that this might be due to latent handling-induced mortality. If true, radio-tagged fish could be affected more severely than fish tagged only with spaghetti tags, and could explain the large proportion of radio-tagged sample remaining in the Yukon Flats. This explanation is supported by the fact that, when combined, abundance estimates from the primary assessment projects in the upper basin (i.e., Chandalar and Sheenjek sonar, Fishing Branch weir, and the Yukon River mark-recapture study in Canada) and mark-recapture estimates for the entire return (Underwood et al. 2000) were comparable in 1998 and 1999, arguing against a large spawning population in the Yukon Flats. Stock abundance estimates in 1999 from the Chandalar River sonar and Yukon River mark-recapture study were also significantly greater than the radio-tagging estimates for these areas (Figure 10).

Several factors, however, argue against the hypothesis that radio-tagged fish died prematurely due to handling. Relatively few fish remained in non-terminal areas in 1998 compared to 1999, and it is

difficult to explain why handling affects would be more severe during the second year of the study. Capture and handling procedures were the same in both years, with the exception that a tissue sample (i.e., fin clip) was taken in 1999 for genetic analysis. In addition, the tagging crew was comprised of the same people in both years, and the care and quality of handling would generally improve with increased experience. Water levels were higher in 1999, based on limited information available for the upper basin (National Weather Service, unpublished data from Eagle, Alaska) which could impact fish in a weakened state if higher water resulted in increased current in areas used by migrating fish. However, water levels in 1999 declined later in the run when increased numbers of fish, tracked to the Yukon Flats, were tagged. Marked-unmarked ratios for radio-tagged fish recovered in fisheries in lower and upper sections of the basin were not significantly different during both 1998 and 1999, although the sample sizes were relatively small. If handling adversely impacted the fish, the percentage affected should be comparable during weekly periods. Instead, the percent of radio-tagged fish remaining in the Yukon Flats increased later in the run (Figure 4). Seasonal changes in river conditions (including water temperature and river stage), physiological conditions of fish later in the run, or a combination of these factors may have contributed. Late-run pink salmon (*O. gorbuscha*) have a shorter redd life than early run fish (Helle et al. 1964), suggesting that late run fish may have lower physical reserves and are more susceptible to other stresses. However, radio-tagged fish returning to the Yukon River in Canada were prevalent later in the run (Figure 4), and were comprised mainly of upper Yukon River fish that spawned in main stem areas.

The decline in marked-unmarked ratios reported for spaghetti-tagged chum salmon also suggests a progressive affect, i.e., fish continually eliminated from the population as they traveled further upriver, rather than tagged fish dying en mass after release. If true for radio-tagged fish, the proportions

observed in non-terminal reaches of the upper basin, such as the lower and upper Porcupine River, should reflect this phenomenon with inflated numbers. Relatively few radio-tagged fish were observed in the lower Porcupine River, and the proportions observed in 1998 and 1999 were comparable (Table 4). Aerial surveys were not conducted in this area, although there are unconfirmed reports of salmon spawning in main stem reaches and associated tributaries. Fish were observed in the upper Porcupine River, although the percentages were again comparable between years. Little information is available on spawning distribution in the Canadian portion of the drainage, other than counts at the Fishing Branch River weir. Aerial surveys conducted in 1998 located over 28% of the fish tracked to this area in upper reaches of the main stem, Miner River, and lower Fishing Branch River (downstream from the weir), while 24% were not located and presumed to be in off-channel areas not surveyed (Boyce and Eiler 2000). The status of the fish located in mainstem areas was not determined due to ice on most sections of the river and limited visibility in reaches that were open. No spawning was observed during aerial surveys conducted in mainstem areas during 2001 (B. Borba, Alaska Department of Fish and Game, personal communication), suggesting that radio-tagged fish located in 1998 either died while in-transit to spawning areas or drifted downstream after spawning in areas further upriver. Radio-tagged fish were located in areas immediately upstream from the upper main stem (i.e., Miner and lower Fishing Branch Rivers), and could likely have drifted downstream, particularly since aerial surveys in 1998 were conducted late in the fall when most fish had died or were in post-spawning condition. Escapement estimates for the Fishing Branch River, calculated using radio-tagged fish tracked to the lower and upper Porcupine River, as well as those actually returning to the Fishing Branch weir (Figure 11), suggest that mortality while in-transit to spawning areas above the weir was minimal.

If handling has a latent affect on a portion of the fish tagged, other fish in the sample, particularly those traveling extended distances, would likely exhibit an intermediate response such as a reduced movement rate. Fish traveling to areas closer to the tagging site would be expected to be less affected. However, movement rates for fish traveling to upper reaches of the basin were comparable or faster than fish tracked to lower reaches such as the Chandalar and Sheenjek Rivers. Fish destined for lower reaches may move slower normally since they have less distance to travel prior to spawning. Proximity to spawning areas may also be a factor (e.g., tracking stations on both the Chandalar and Sheenjek Rivers were located relatively close to known spawning areas), although movement rates for Fishing Branch River fish (38.5 km/day in 1998 and 35.7 km/day in 1999) were similar to other upriver stocks even though this tracking station was located near spawning areas. Movement rates were more variable for females tracked to the Porcupine River drainage, although this was due primarily to variation between the different areas (i.e., movement rates for lower Porcupine River females were comparable to Sheenjek and Chandalar River females) and movements from other reaches of the basin (e.g., fish that originally traveled to the Sheenjek River prior to moving to the lower Porcupine).

Movement rates exhibited by radio-tagged fish were consistently less in 1999 than in the previous year, possibly in response to higher water levels. The pattern was consistent between sexes and for fish returning to different reaches of the basin. However, the difference between years for Porcupine River fish was significantly less than observed for other stocks (Figure 7), and this observation would not be expected if handling was impacting performance in relation to distance traveled, since Porcupine River fish traveled substantially farther after release (1252 km) than fish returning to the Chandalar (419 km) and Sheenjek (502 km) Rivers. However, other factors may be involved. Water

levels, which could potentially impact swimming performance, on the Porcupine and Yukon Rivers can differ within a given year, and records for the Porcupine River are not available. Qualitative comments from local fishers indicated that chum salmon harvested in 1999 were in better condition than the previous year, suggesting that the lower movement rates were unrelated to the general fitness of the fish. Milligan et al. (1986) reported that radio-tagged chum salmon, tagged above the U.S.-Canada border in 1982, traveled an average of 37.9 km/day, a rate comparable to fish radio tagged at the Rapids and tracked to the upper Yukon River in 1999 (i.e., 36.5 km/day). This comparison suggests that the fish in 1999 had normal patterns of movement after tagging compared to fish in the 1982 study, despite the additional distance traveled (i.e., 890 km from the Rapids tagging site to the U.S.-Canada border). The limited information available also suggests that the migration rates observed during our study were comparable or greater than reported for other river drainages, although the extreme distances traveled by Yukon River chum salmon during inriver spawning migrations make the significance of these comparisons uncertain in terms of assessing potential tagging affects.

Most Yukon Flats fish were not located during fall aerial surveys in 1999, especially fish tagged later in the run (Figure 4). This was particularly interesting since most of the main river was surveyed, suggesting that the fish were either in off-channel areas (i.e., side channels, sloughes, and secondary tributaries) not surveyed, were not detected during surveys due to equipment problems, or had left the Yukon Flats without being detected. If radio-tagged fish succumbed to excessive stress or injuries from handling and tagging, they may have drifted downriver and out of the area. However fish drifting with the current past the Rapids tracking stations would probably have been recorded, since they would be within reception range for an extended period of time, increasing the chance of being detected. Fish that died and sank into deep water would be difficult to detect. However water depths

in the main river and associated side channels were thought to be relatively shallow due to the braided nature of the Yukon Flats and the lower water levels in the fall, making it unlikely that radio tags were not detected in the areas surveyed. Although aerial tracks were conducted in the Yukon Flats during the fall of 1999, the entire area could not be covered thoroughly, and groups of fish may have gone undetected in side channels and sloughs associated with the main river or secondary tributaries not surveyed. Equipment problems are another possible explanation. Transmitter failures are known to occur, although failure rates are usually minor. All the tags used during this study were tested prior to deployment, although this screening may not have been sufficient to detect problems that developed over time. Although tracking station performance was generally good during the study, technical problems periodically occurred with the receiving equipment. Output from the station provided by the satellite uplink was monitored daily to evaluate station performance, and sites visited when problems occurred. Station records from the U.S.-Canada border indicated that technical problems occurred when fish from Week 38 were traveling through the area, and the percentage of upper Yukon River fish during this period was underestimated. Station performance prior to detecting the problem was possibly substandard, resulting in radio-tagged fish passing the site undetected, and further underestimating this component of the return.

One of the strongest arguments for fish being impaired by handling is that radio tagging stock abundance estimates were substantially less in 1999 than estimates from the Chandalar River sonar and the Yukon-River mark-recapture study in Canada (Figure 10). However, uncertainties also exist related to these estimates. Sonar on the right bank of the Chandalar River, which is assumed to represent 57% of the count, was inoperable for 28 days in 1999 due to high water, although the left bank sonar only missed a few days, and the relationship between right and left bank counts was highly

correlated when water levels were lower. This data gap is particularly inopportune since most chum salmon in 1999 returned during the early portion of the run (Figure 9). All of the Chandalar River fish radio tagged during Weeks 32-34, and most in Week 35, passed the Chandalar River sonar site during this period of high water. Limited information is available on the pattern of movement exhibited by radio-tagged fish passing the sonar site due to the resolution of the tracking station data. Most fish appeared to move directly past the site within one to three hours of being detected, although some fish made distinct movements back downriver, before passing upriver a final time. Sonar data indicate that most fish moving upriver were highly bank oriented, whereas fish moving downriver were evenly distributed across the ensonified zone (Daum and Osborne 2000), and the extent that fish move downriver through areas not covered by sonar, prior to resuming movements upriver, is not known. Questions also exist related to mark-recapture estimates from Yukon River tagging studies in Canada. In addition to sample size limitations, sampling bias associated with gear selectivity and changes in river conditions are widely recognized. Handling methods may also have an impact, since fish marked with spaghetti tags are held in fish wheel live boxes for varying periods of time. Milligan et al. (1986) reported 11% handling mortality for chum salmon radio tagged in the Yukon River in Canada, although a portion of these fish may have regurgitated their tags. Similar tagging mortality demonstrated by spaghetti-tagged fish would bias abundance estimates. Handling effects could have been more severe for radio-tagged fish, however, radio-tagged fish were not held in fish wheel live boxes prior to tagging and release, which may have reduced anomalous behavior. Stock abundance estimates based on the radio-tagged fish higher than Sheenjek River sonar estimates and comparable to weir counts at the Fishing Branch weir (Figure 10) which would not be expected if radio-tagged fish moving further upriver were experiencing higher levels of mortality. In particular, differences at the Fishing Branch weir would likely be more distinct, considering the distances traveled (over 1250 km),

if this was a factor. In conclusion, the information currently available does not fully support any of the suggested explanations, and additional research is needed to develop a better understanding of the factors involved. An analysis of the genetic information collected during the study may also provide insights useful in addressing some of the questions raised.

Effects of Holding Fish

Although initially developed as a harvest method, fish wheels have been used to study salmon returns in numerous rivers along the western coast of North America (Milligan et al. 1986, McGregor et al. 1991, Link et al. 1996), providing an efficient capture method for enumeration, tagging and sampling projects. Although test fisheries have traditionally been used to assess run abundance in large rivers based on catch per unit effort estimates, fish wheels equipped with live boxes provide a non-lethal alternative for collecting comparable information (Fliris 2001). Live boxes are routinely used to hold fish until they are processed, reducing the time personnel need to be present at the site. A primary assumption with this approach is that the fish captured and released are not adversely affected by the process.

The declining marked-unmarked ratios reported by Underwood et al. (2002), suggest that chum salmon tagged at the Rapids were experiencing tagging induced mortality as they continued their upriver migration. However, fish radio tagged at the Rapids in 1998 did not exhibit a negative response, based on the movements and distribution observed, which suggests that other factors are involved. Fish tagged as part of the mark-recapture study were held in fish wheel live boxes for varying periods of time prior to being tagged and released, while radio-tagged fish were tagged and released immediately after capture. Numerous studies have reported that holding causes stress

(Robertson et al. 1987, Mazur and Iwama 1993), although the impacts on performance and survival are often variable and difficult to assess. Radio telemetry studies on salmon in the Taku River (Eiler et al. 1991, Eiler 1995) stopped tagging fish held in live boxes when significant differences in response were observed between held and immediately released fish. Adverse affects were most evident for chinook salmon, with held fish drifting downstream to salt water after release, a distance of over 20 km, prior to resuming upriver movement – a response not exhibited by immediately released fish.

Results from the present study demonstrate that the movement patterns displayed by chum salmon radio tagged at the Rapids differed for held and immediately released fish. The percentage of held fish that resumed upriver movements, although comparable, was consistently lower during the three years of the study. Held fish took significantly longer to resume upriver movements than immediately released fish. If handling had no effect, fish released at the Rapids should have traveled upriver past the Rapids tracking station in approximately 7-11 hours (0.28-0.33 days) based on the average movement rates of all radio-tagged fish that moved upriver. Instead, average travel times to the Rapids station for immediately release fish ranged between 0.51 day and 0.99 day compared to 2.02 days to 2.85 days for held fish, suggesting that both handling procedures had an initial affect, although the delay exhibited by held fish was more severe. As previously mentioned, fish will often exhibit an initial delay after being handled, however, long-term differences were also observed between the two groups. Distribution information from the holding experiments indicated that held fish responded differently than immediately released fish, with fewer immediately released fish remaining in the Yukon Flats (2% in 1998, 19% in 1999) compared to held fish (24 % in 1998, 57% in 1999). Similar results were also observed for fish recaptured and released at research fish wheels at Rampart. In

addition, most fish observed drifting downriver past the Rapids tracking stations had been held in fish wheel live boxes, particularly those recaptured by fish wheels at Rampart. The responses by held fish were also more variable than those for immediately released fish. For example, although the average time taken to resume upriver movements was significantly greater for held fish, some of the fish behaved like the immediately released fish. This result supports the contention that stress was a contributing factor, since individuals often vary in their response to adverse conditions. Potentially, affected individuals could exhibit a progressive pattern of impacted behavior similar to that suggested by Underwood et al. (2002).

Concerns related to holding fish are not universal, even within the Yukon River basin. Mark-recapture studies on chum salmon in the Tanana River, where fish are held in fish wheel live boxes prior to tagging, have not observed reduced marked-unmarked ratio in recovery projects further upriver (B. Borba, Alaska Department of Fish and Game, personal communication). Although our holding tests attempted to control certain conditions, fish held in live boxes in most field studies may be influenced by a number of variables including the length of time held (e.g., up to 12 hours in some projects), the number of fish confined, configuration of the live box, the physiological condition of the fish, physical characteristics of the river (e.g., water temperature and velocity), distance to spawning areas, or a combination of these and other factors. Fish wheels are a valuable management tool, providing an effective, and presumably non-lethal, method for capturing large numbers of fish. In light of the expanding role of fish wheels in research and run assessment programs, a better understanding of the potential impacts of this method on fish resources is needed.

Acknowledgments

This study was part of a cooperative program conducted with the U.S. Fish and Wildlife Service, Fairbanks Fisheries Research Office. Support for the program was also provided by the Alaska Department of Fish and Game, Department of Fisheries and Oceans Canada, Council of Athabaskan Tribal Governments, and the Chalkyitsik Village Council. We thank I. Boyce, B. Carter, P. Erhart, S. Ho, R. Holder, J. Jozefech, J. Melegari, R. Morris, and J. Tilly for assistance with field work. Stan Zuray maintained and operated the fish wheels at the tagging site. J. Helle and R. Wilmot critically review the manuscript. We are particularly grateful to M. Millard, J. Greenough, and M. Dahlberg for their encouragement and support throughout the study.

Literature Cited

- Anonymous. 1985. Report on the status of salmon stocks, fisheries, and management programs in the Yukon River. Unpublished report prepared by the scientific working group for the delegations from United States and Canada concerning the Yukon River, April 23-24, 1985. Anchorage.
- Barton, L. H. 1992. Tanana River, Alaska, fall chum salmon radio telemetry study. Alaska Department of Fish and Game, Division of Commercial Fisheries. Fishery Research Bulletin No. 92-01. Juneau, Alaska.
- Boyce, I., and J. H. Eiler. 2000. Distribution of chum salmon in the Porcupine River drainage, Canada. Pages 51-58 *in* J. H. Eiler, D. J. Alcorn, and M. R. Neuman (editors). Biotelemetry 15: Proceeding of the 15th International Symposium on Biotelemetry. Juneau, Alaska USA. International Society on Biotelemetry. Wageningen, The Netherlands.
- Brown, R. J. 2000. Migratory patterns of Yukon River Inconnu as determined with otolith microchemistry. Masters Thesis. University of Alaska, Fairbanks, Alaska. 80 pp.
- Brown, R. J., and J. H. Eiler. 2000. Performance of Yukon River Inconnu tagged with external and internal radio transmitters. Pages 145-153 *in* J. H. Eiler, D. J. Alcorn, and M. R. Neuman (editors). Biotelemetry 15: Proceeding of the 15th International Symposium on Biotelemetry. Juneau, Alaska USA. International Society on Biotelemetry. Wageningen, The Netherlands.

Daum, D. W., and B. M. Osborne. 2000. Enumeration of Chandalar River fall chum salmon using split-beam sonar, 1999. U. S. Fish and Wildlife Service, Fairbanks Fishery Resource Office, Alaska Fisheries Technical Report Number 58, Fairbanks, Alaska. 37 pp.

Eiler, J. H. 1990. Radio transmitters used to study salmon in glacial rivers. *American Fisheries Society Symposium* 7: 364-369.

Eiler, J. H., B. D. Nelson and R. F. Bradshaw. 1991. Radio tracking chinook salmon (*Oncorhynchus tshawytscha*) in a large turbid river. Pages 202-206 *in* A. Uchiyama and C. J. Amlaner, Jr. (editors). *Biotelemetry 11: Proceeding of the 11th International Symposium on Biotelemetry*. Waseda University Press. Yokohama, Japan.

Eiler, J. H., B. D. Nelson, and R. F. Bradshaw. 1992. Riverine spawning by sockeye salmon in the Taku River, Alaska and British Columbia. *Transactions of the American Fisheries Society* 121:701-708.

Eiler, J. H. 1995. A remote satellite-linked tracking system for studying Pacific salmon with radio telemetry. *Transactions of the American Fisheries Society* 124:184-193.

Eiler, J. H. and M. A. Masters. 2000. A database-GIS mapping program for summarizing salmon telemetry data from the Yukon River basin. Pages 138-144 *in* J. H. Eiler, D. J. Alcorn, and M. R. Neuman (editors). Biotelemetry 15: Proceeding of the 15th International Symposium on Biotelemetry. Juneau, Alaska USA. International Society on Biotelemetry. Wageningen, The Netherlands.

Fliris, B., 2001. Modifications of video storage equipment for purposes of providing accurate catch-per-unit-effort data from the Yukon River Subdistrict 5-A test fishwheel. Yukon River Panel Final Report. Yukon River Panel. Whitehorse, Yukon Territory, Canada. 15 pp.

Gilbert, C. H. 1922. The salmon of the Yukon River. U.S. Bureau of Fisheries Bulletin 38:317-332.

Goodman, L. A. 1960. On the exact variance of products. Journal of the American Statistical Association 55: 708-713. (Corrected in Journal of the American Statistical Association 57: 917).

Gordon, J. A., S. P. Klosiewski, T. J. Underwood, and R. J. Brown. 1998. Estimated abundance of adult fall chum salmon in the upper Yukon River, Alaska, 1996. U. S. Fish and Wildlife Service, Fairbanks Fishery Resource Office, Alaska Fisheries Technical Report Number 45, Fairbanks, Alaska. 43 pp.

- Greenough, J. W. 1971. Estimation of sockeye, coho, and chinook salmon runs at Wood Canyon on the Copper River in 1966, 1967, and 1968. U.S. Department of Interior, Fish and Wildlife Service Fishery Resource Report on the Wood Canyon Dam Project. Juneau, Alaska. 109 pp.
- Haegele, C. W. 1990. Anchor tag return rates for Pacific herring in British Columbia. American Fisheries Society Symposium 7:127-133.
- Helle, J. H., and M. S. Hoffman. 1995. Size decline and older age at maturity of two chum salmon (*Oncorhynchus keta*) stocks in western North America, 1972-92. Pages 245-260 in R. J. Beamish (editor). Climate Change and Northern Fish Populations. Canadian Special Publication of Fisheries and Aquatic Sciences 121.
- Helle, J. H., R. S. Williamson, and J. E. Bailey. 1964. Intertidal ecology and life history of pink salmon at Olsen Creek, Prince William Sound, Alaska. U.S. Fish and Wildlife Service Special Scientific Report - Fisheries No. 483:1-19.
- Kelsch, S. W., and B. Shields. 1996. Care and handling of sampled organisms. Pages 121-144 in B. R. Murphy and D. W. Willis (editors). Fisheries Techniques, 2nd edition. American Fisheries Society. Bethesda, MD.
- Link, M. R., K. K. English, and R. C. Bocking. 1996. The 1993 fishwheel project on the Nass River and an evaluation of fishwheels as an inseason management and stock assessment tool for the Nass River. Canadian Manuscript Report of Fisheries and Aquatic Sciences No. 2372, 92 pp.

- Matthews, K. R., and R. H. Reavis. 1990. Underwater tagging and visual recapture as a technique for studying movement patterns of rockfish. *American Fisheries Society Symposium* 7:168-172.
- Mazur, C. F., and G. K. Iwama. 1993. Handling and crowding stress reduces number of plaque-forming cells in Atlantic salmon. *Journal of Aquatic Animal Health* 5:98-101.
- McCleave, J. D., and K. A. Stred. 1975. Effect of dummy telemetry transmitters on stamina of Atlantic salmon (*Salmo salar*) smolts. *Journal of the Fisheries Research Board of Canada* 32:559-563.
- McGregor, A. J., P. A. Milligan and J.E. Clark. 1991. Adult mark-recapture studies of Taku River salmon stocks in 1989. Alaska Department of Fish and Game Division of Commercial Fisheries Technical Fishery Report 91-05. Alaska Department of Fish and Game, Juneau, Alaska. 73 pp.
- Meehan, W. R. 1960. Use of a fish wheel in salmon research and management. *Transactions of the American Fisheries Society* 90:490-494.
- Mellas, E. J., and J. M. Haynes. 1985. Swimming performance and behavior of rainbow trout (*Salmo gairdneri*) and white perch (*Morone americana*): effects of attaching telemetry transmitters. *Canadian Journal of Fisheries and Aquatic Science* 42:488-493.

- Milligan, P. A., W. O. Rublee, D. D. Cornett and R. A. C. Johnston. 1986. The distribution and abundance of chum salmon (*Oncorhynchus keta*) in the upper Yukon River basin as determined by a radio-tagging and spaghetti tagging program: 1982-1983. Canadian Technical Report of Fisheries and Aquatic Sciences No. 1351. 141 pp.
- Pella, J. J., J. C. Olsen, J. H. Helle, J. H. Eiler, J. A. Orsi, and D. G. Mortensen. 1988. Sockeye salmon (*Oncorhynchus nerka*) escapements to southern Southeast Alaska in 1982 and 1983. National Oceanic and Atmospheric Administration Technical Memorandum, National Marine Fisheries Service F/NWC-150.
- Raymond, M. L. and F. Rousset. 1995. An exact test for population differentiation. *Evolution* 49:1280-1283.
- Robertson, L. P., C. R. Arnold, and J. M. Trant. 1987. Plasma cortisol and secondary stress responses of red drum to handling, transport, rearing density and disease outbreak. *Progressive Fish-Culturist* 49:1-12.
- Sokal, R. R. and F. J. Rohlf. 1969. *Biometry: The Principles and Practice of Statistics in Biological Research*. W. H. Freeman and Company, New York. 776 pp.
- Salo, E. O. 1991. Life history of chum salmon (*Oncorhynchus keta*). Pages 231-310 in C. Groot and L. Margolis (editors). *Pacific Salmon Life Histories*. University of British Columbia Press, Vancouver, British Columbia.

Underwood, T. J., S. P. Klosiewski, J. A. Gordon, J. L. Melegari, and R. J. Brown. 2000a. Estimated abundance of adult fall chum salmon in the middle Yukon River, Alaska, 1997. U. S. Fish and Wildlife Service, Fairbanks Fishery Resource Office, Alaska Fisheries Technical Report Number 56, Fairbanks, Alaska. 48 pp.

Underwood, T. J., S. P. Klosiewski, J. L. Melegari, and R. J. Brown. 2000b. Estimated abundance of adult fall chum salmon in the middle Yukon River, Alaska, 1998-1999. U. S. Fish and Wildlife Service, Fairbanks Fishery Resource Office, Alaska Fisheries Technical Report Number 57, Fairbanks, Alaska. 29 pp.

Underwood, T. J., J. F. Bromaghin, and S. P. Klosiewski. 2002 Evidence of handling mortality in fall chum salmon caused by fish wheel capture on the Yukon River, Alaska. U.S. Fish and Wildlife Service, Fairbanks Fishery Resource Office, Alaska Fisheries Technical Report Number 59, Fairbanks, Alaska.

Wilmot, R. L., R. Everett, W. J. Spearman, and R. Baccus. 1992. Genetic stock identification of Yukon River chum and chinook salmon. U. S. Fish and Wildlife Service Progress Report. U. S. Fish and Wildlife Service. Anchorage, Alaska. 132 pp.

Wydoski, R. and L. Emery. 1983. Tagging and marking. Pages 215-237 in L. A. Nielsen and D. L. Johnson (editors). Fisheries Techniques. American Fisheries Society. Bethesda, MD.

Yukon River Joint Technical Committee. 2000. Yukon River Salmon Season Review for 2000 and Technical Committee Report. U.S./Canada Yukon River Joint Technical Committee Report. Anchorage, Alaska. 67 pp.

Zuray, S., 2001. Rampart-Rapids summer catch-per-unit-effort video monitoring - 2001. Federal Subsistence Fishery Monitoring Program Annual Report. U.S. Fish and Wildlife Service. Anchorage, Alaska. 30 pp.